

WORKEDOUT EXAMPLES

W. E -1 : Verify whether $f(x) = x^{1/5}$ has a vertical tangent at $x = 0$

Sol: At $x = 0$, $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x)^{1/5} = 0$

And $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x)^{1/5} = 0$

Also $f(0) = 0$

Thus $f(x)$ is continuous at $x=0$

$$\text{Now L.H.D} = \lim_{h \rightarrow 0^+} \frac{f(-h) - 0}{-h} = \lim_{h \rightarrow 0^+} \frac{(-h)^{1/5}}{-h}$$

$$= \lim_{h \rightarrow 0^+} \frac{1}{(-h)^{4/5}} = +\infty$$

$$\begin{aligned}\text{And R.H.D} &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h}\end{aligned}$$

$$= \lim_{h \rightarrow 0^+} \frac{(h)^{1/5}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{(h)^{4/5}} = \infty$$

$\therefore f(x) = x^{1/5}$ has a vertical tangent at $x = 0$.

W.E-2 : Consider the function $f(x)$ defined by

$$f(x) = \begin{cases} x \sin \frac{1}{x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$$

$$\sin \frac{1}{x} \in [-1, 1] \quad (x \neq 0)$$

then $x \sin \frac{1}{x}$ lies in between $y = -x$ and $y = x$

and $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 \times (\text{a finite real number}) = 0 = f(0)$

Thus $f(x)$ is continuous at $x=0$

Now

$$\text{L.H.D} = \lim_{h \rightarrow 0^+} \frac{f(-h) - 0}{-h} = \lim_{h \rightarrow 0} \left(-\sin \frac{1}{h} \right)$$

which oscillates in between -1 and 1

Similarly R.H.D = $\lim_{x \rightarrow 0} \left(\sin \frac{1}{x} \right)$ which also oscillate in

between -1 and 1

Thus L.H.D and R.H.D do not exist at $x = 0$.

W.E-3. If $f(x) = \begin{cases} |x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$ then

$$\lim_{x \rightarrow 0} (f(f(f(\dots f(x)))) \text{ (where 'f' is taken 'n' times)})$$

- a) 0 b) 1
c) -1 d) Does not exist

Sol: **(D)** $\lim_{x \rightarrow 0} \frac{f(f(f(f \dots f(x))))}{n \text{ times}} = f(x)$

$$\therefore \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{|x|}{x} = \text{does not exist}$$

W. E- 4. If $f(x) = x \sin \pi x$, $x > 0$. Then, for all natural numbers n , $f'(x)$ vanishes at

- (a) a unique point in the interval $\left(n, n + \frac{1}{2}\right)$
- (b) a unique point in the interval $\left(n + \frac{1}{2}, n + 1\right)$
- (c) a unique point in the interval $(n, n + 1)$
- (d) two points in the interval $(n, n + 1)$

Sol: Ans: (c)

$$f'(x) = \sin \pi x + \pi x \cos \pi x = 0 \Rightarrow \tan \pi x = -\pi x$$

$$\Rightarrow \pi x \in \left[\frac{2n+1}{2} \pi, (n+1) \pi \right]$$

$$\Rightarrow x \in \left(n + \frac{1}{2}, n+1\right) \in (n, n+1)$$

W. E- 5. Given

$$f(x) = \begin{cases} 3 - \left[\cot^{-1} \left(\frac{2x^3 - 3}{x^2} \right) \right] & \text{for } x > 0 \\ \{x^2\} \cos(e^{1/x}) & \text{for } x < 0 \end{cases},$$

where $\{\cdot\}$ and $[\cdot]$ denotes the fractional part and the integral part functions respectively, then which of the following statement does not hold good?

- a)** $f(0^-) = 0$

$$\text{b) } f(0^+) = 3$$

$$\text{c) } f(0) = 0 \Rightarrow \text{continuity of } f \text{ at } x = 0$$

$$\text{d) irremovable discontinuity of } f \text{ at } x = 0.$$

Sol: (b, d)

$$f(0^+) = \lim_{h \rightarrow 0} f(0+h)$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} 3 \left[\cot^{-1} \left(\frac{2h^3 - 3}{h^2} \right) \right] \\ &= 3 - \left[\cot^{-1}(-\infty) \right] \\ &= 3 - \left[\pi - \cot^{-1} \infty \right] \\ &= 3 - [\pi - 0] \\ &= 3 - 3 = 0 \\ \therefore f(0^+) &\neq 3 \end{aligned}$$

$$\begin{aligned} f(0^-) &= \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \{h^2\} \cos(e^{-1/h}) \\ &= 0 \times \cos(e^{-\infty}) \\ &= 0 \times \cos(0) \\ &= 0 \times 1 = 0 \end{aligned}$$

$$\therefore f(0^+) = f(0^-) = f(0) = 0$$

$$\therefore f(x) \text{ is continuous at } x = 0$$

W.E-6. If $\lim_{m \rightarrow \infty} \left(\lim_{n \rightarrow \infty} \sum_{k=1}^m \frac{\sqrt[n]{(k-1)^n + k^n}}{m^2} \right) = p$

$$\text{and } \lim_{x \rightarrow -\infty} \left\{ x + \sqrt{x^2 + 5x \cos \frac{1}{|x|}} \right\} = q \text{ then } p - q =$$

Sol: 3

$$\begin{aligned} P &= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{\left(1 + \sqrt[n]{1^n + 2^n} + \sqrt[n]{2^n + 3^n} + \dots + \sqrt[n]{(n-1)^n + n^n} \right)}{m^2} \\ &= \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{\left(1 + 2\sqrt[n]{1 + \left(\frac{1}{2}\right)^n} + 3\sqrt[n]{1 + \left(\frac{2}{3}\right)^n} + \dots + m\sqrt[n]{1 + \left(\frac{m-1}{m}\right)^n} \right)}{m^2} \end{aligned}$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$= \lim_{m \rightarrow \infty} \left(\frac{1 + 2 + 3 + \dots + m}{m^2} \right)$$

$$= \lim_{m \rightarrow \infty} \frac{m(m+1)}{2m^2} = \frac{1}{2}$$

$$q = \lim_{x \rightarrow -\infty} \left\{ x + \sqrt{x^2 + 5x \cos \frac{1}{x}} \right\}$$

$$= \lim_{x \rightarrow -\infty} \left(\frac{-5x \cos \frac{1}{x}}{x - \sqrt{x^2 + 5x \cos \frac{1}{x}}} \right)$$

$$= \lim_{x \rightarrow -\infty} \left(\frac{-5x \cos \frac{1}{x}}{1 + \sqrt{1 + \frac{5}{x} \cos \frac{1}{x}}} \right) = \frac{-5}{2}$$

$$\therefore p - q = 3$$

W.E-7. If $f(n, \theta) = \prod_{r=1}^n \left(1 - \tan^2 \frac{\theta}{2^r} \right)$ then

$$\lim_{\theta \rightarrow 0} \left(\lim_{n \rightarrow \infty} f(n, \theta) \right) =$$

Sol: (1) $f(n, \theta) = \prod_{r=1}^n \left(1 - \tan^2 \frac{\theta}{2^r} \right)$

$$= \frac{2 \tan \frac{\theta}{2}}{\tan \theta} \cdot \frac{2 \tan \frac{\theta}{2^2}}{\tan \frac{\theta}{2}} \cdot \frac{2 \tan \frac{\theta}{2^3}}{\tan \frac{\theta}{2^2}} \dots$$

$$f(n, \theta) = \frac{2^n \tan \frac{\theta}{2^n}}{\tan \theta} \Rightarrow \lim_{n \rightarrow \infty} f(n, \theta)$$

$$= \lim_{n \rightarrow \infty} \frac{\theta \tan \frac{\theta}{2^n}}{(\tan \theta) \frac{\theta}{2^n}} = \frac{\theta}{\tan \theta}$$

$$\lim_{\theta \rightarrow 0} \left(\frac{\theta}{\tan \theta} \right) = 1$$

W.E-8 Let $x_0 = 2 \cos \frac{\pi}{6}$ and $x_n = \sqrt{2 + x_{n-1}}$,

$n=1, 2, 3, \dots$ then $\left[\lim_{n \rightarrow \infty} 2^{(n+1)} \sqrt{2 - x_n} \right]$ (where $[.]$ is G.I.F)

Sol: (1) $x_1 = \sqrt{2 + 2\cos\frac{\pi}{6}} = 2\cos\frac{\theta}{2}$ where $\theta = \frac{\pi}{6}$

$$x_2 = 2\cos\frac{\theta}{4}, \dots, x_n = 2\cos\frac{\theta}{2^n}$$

$$\therefore \lim_{n \rightarrow \infty} 2^{n+1} \sqrt{2 - 2\cos\frac{\theta}{2^n}}$$

$$= \lim_{n \rightarrow \infty} 2^{n+1} 2\sin\frac{\theta}{2^{n+1}}$$

$$= 2\theta = 2 \times \frac{\pi}{6} = \frac{\pi}{3}$$

$$\therefore \left[\frac{\pi}{3}\right] = 1$$

LEVEL - V

SINGLE ANSWER QUESTIONS

1. The integral value of n for which

$$\lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x - e^x \cos x + e^x - \left(\frac{x^3}{2}\right)}{x^n} \text{ is}$$

finite and non zero is

- (A) 2 (B) 4 (C) 5 (D) 6

2. The value of

$$\lim_{x \rightarrow \infty} \frac{2(x)^{1/2} + 3(x)^{1/3} + 4(x)^{1/4} + \dots + n(x)^{1/n}}{(2x-3)^{1/2} + (2x-3)^{1/3} + \dots + (2x-3)^{1/n}} \text{ is}$$

- (A) $\sqrt{2}$ (B) 2 (C) $\frac{1}{\sqrt{3}}$ (D) 0

3. ABC is an isosceles triangle inscribed in a circle of radius r . If $AB = AC$ and h is the altitude from A to BC , then the triangle ABC has perimeter $P =$

$$2\left[\sqrt{(2hr - h^2)} + \sqrt{2hr}\right] \text{ and area } A$$

then $\lim_{h \rightarrow 0} \frac{A}{P^3}$

- (A) $\frac{1}{r}$ (B) $\frac{1}{64r}$ (C) $\frac{1}{128r}$ (D) $\frac{1}{2r}$

4. The integer n for which

$$\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n} \text{ is a finite non-}$$

zero number is

[IIT - 2002]

- (A) 1 (B) 2
(C) 3 (D) 4

5. If $A_i = \frac{x - a_i}{|x - a_i|}$, $i = 1, 2, 3, \dots, n$ and

$a_1 < a_2 < a_3 < \dots < a_n$, then

$$\lim_{x \rightarrow a_m} (A_1 A_2 \dots A_n), 1 \leq m \leq n$$

(A) is equal to $(-1)^m$ (B) is equal to $(-1)^{m+1}$

(C) is equal to $(-1)^{m-1}$ (D) Does not exist

6. The value of $\lim_{x \rightarrow \infty} \left\{ \frac{x}{x + \frac{\sqrt[3]{x}}{x + \frac{\sqrt[3]{x}}{x + \frac{\sqrt[3]{x}}{\dots \infty}}}} \right\}$ is

- (A) 1 (B) 0 (C) 2 (D) $\frac{1}{2}$

7. If $\lim_{x \rightarrow 0} \left[\left(1 - \sqrt{\cos 2x} \sqrt[3]{\cos 3x} \sqrt[4]{\cos 4x} \dots \sqrt[n]{\cos nx} \right) / x^2 \right]$

is equal to 10, then the value of n is

- (A) 5 (B) 4 (C) 6 (D) 3

8. Let $a_1, a_2, \dots, a_n$ be sequence of real

numbers with $a_{n+1} = a_n + \sqrt{1 + a_n^2}$ and

$$a_1 = 0. \text{ Then } \lim_{n \rightarrow \infty} \left(\frac{a_n}{2^{n-1}} \right) =$$

- (A) $\frac{2}{\pi}$ (B) $-\frac{\pi}{2}$ (C) $-\frac{2}{\pi}$ (D) $\frac{\pi}{2}$

9. If $\lim_{x \rightarrow 0} (x^{-3} \sin 3x + ax^{-2} + b)$ exists and is equal to 0, then

(A) $a = -3$ and $b = \frac{9}{2}$

(B) $a = 3$ and $b = \frac{9}{2}$

(C) $a = -3$ and $b = -\frac{9}{2}$

(D) $a = 3$ and $b = -\frac{9}{2}$

10. If $\lim_{x \rightarrow 0} [1 + x \ln(1 + b^2)]^{1/x} = 2b \sin^2 \theta, b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is [IIT-2011]

(A) $\pm \frac{\pi}{4}$ (B) $\pm \frac{\pi}{3}$ (C) $\pm \frac{\pi}{6}$ (D) $\pm \frac{\pi}{2}$

11. If $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$ then, [IIT-2012]
- (A) $a = 1, b = 4$ (B) $a = 1, b = -4$
(C) $a = 2, b = -3$ (D) $a = 2, b = 3$

12. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation

$$(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$$

where $a > -1$. The $\lim_{a \rightarrow 0^+} \alpha(a)$ and

$\lim_{a \rightarrow 0^+} \beta(a)$ are [IIT-2012]

(A) $-\frac{5}{2}$ and 1 (B) $-\frac{1}{2}$ and -1
(C) $-\frac{7}{2}$ and (D) $-\frac{9}{2}$ and 3

13. Which of the following is differentiable at $x = 0$? [IIT - 2000]

(A) $\cos(|x|) + |x|$ (B) $\cos(|x|) - |x|$
(C) $\sin(|x|) + |x|$ (D) $\sin(|x|) - |x|$

14. For the function $f(x) = \begin{cases} \frac{x}{1 + e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$,

the derivative from the right,

$f'(0+)$ and the derivative from

the left, $f'(0-)$ are

(A) $\{0, 1\}$ (B) $\{1, 0\}$ (C) $\{1, 1\}$ (D) $\{0, 0\}$

15. Let $f: [0, \frac{\pi}{2}] \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \max \left\{ \sin x, \cos x, \frac{3}{4} \right\}, \text{ then number}$$

of points where $f(x)$ is non differentiable is
(A) 1 (B) 2 (C) 3 (D) 0

16. Let $f(x) = \begin{cases} \frac{1}{|x|} & \text{for } |x| \geq 1 \\ ax^2 + b & \text{for } |x| < 1 \end{cases}$. If $f(x)$ is continuous and differentiable everywhere, then

(A) $a = \frac{1}{2}, b = -\frac{3}{2}$ (B) $a = -\frac{1}{2}, b = \frac{3}{2}$
(C) $a = 1, b = -1$ (D) $a = b = 1$

17. Let $f(x) = [3 + 2 \cos x], x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

where $[.]$ denotes the greatest integer function. Then number of points of

discontinuity of $f(x)$ is

(A) 3 (B) 2 (C) 5 (D) 6

18. Let $f(x)$ be a real function not identically zero in $\mathbb{R}$, such that

$$f(x + y^{2n+1}) = f(x) + \{f(y)\}^{2n+1}, n \in \mathbb{N} \text{ and}$$

$x, y \in \mathbb{R}$. If $f'(0) \geq 0$, then $f'(6)$ is equal to

(A) 0 (B) 1 (C) -1 (D) 2

19. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the equation $f(x)f(y) - f(xy) = x + y \forall x, y \in \mathbb{R}$ and $f(1) > 0$, then

(A) $f(x)f^{-1}(x) = x^2 - 4$

(B) $f(x)f^{-1}(x) = x^2 - 6$

(C) $f(x)f^{-1}(x) = x^2 - 1$

(D) $f(x)f^{-1}(x) = x^2 + 6$

20. If $f(x) = \frac{x^2 - 1}{x^2 + 1}$, for every real number x , then the minimum value of [IIT - 1998]

(A) does not exist because f is unbounded

(B) is not attained even though f is bounded

(C) is equal to 1 (D) is equal to -1

21. The function $f(x) = [x]^2 - [x^2]$ (where $[x]$ is the greatest integer less than or equal to x), is discontinuous at: [IIT - 1999]

(A) all integers

(B) all integers except 0 and 1

(C) all integers except 0

(D) all integers except 1

22. (i) The left hand derivative of,

$f(x) = [x] \sin(\pi x)$ at $x = k$, k an integer ($[.]$ denotes G.I.F) is [IIT - 2000]

- (A) $(-1)^k (k-1)\pi$ (B) $(-1)^{k-1} (k-1)\pi$
(C) $(-1)^k k\pi$ (D) $(-1)^{k-1} k\pi$

23. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be any function. Define

$g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = |f(x)|$ for all x .

Then g is [IIT - 2000]

- (A) onto if f is onto
(B) one - one if f is one - one
(C) continuous if f is continuous
(D) differentiable if f is differentiable

24. If $f(x) = \frac{x}{1+x} + \frac{x}{(x+1)(2x+1)} + \frac{x}{(2x+1)(3x+1)} + \dots$,

then at $x = 0$, $f(x)$

- (A) has no limit
(B) is discontinuous
(C) is continuous but not differentiable
(D) is differentiable

25. The domain of the derivative of the

function $f(x) = \begin{cases} \tan^{-1} x & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1) & \text{if } |x| > 1 \end{cases}$ is

[IIT - 2002]

- (A) $\mathbb{R} - \{0\}$ (B) $\mathbb{R} - \{1\}$
(C) $\mathbb{R} - \{-1\}$ (D) $\mathbb{R} - \{-1, 1\}$

26. Let $f(x)$ be a strictly increasing and differentiable function, then

$\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$, equals [IIT - 2004]

- (A) 2 (B) 1 (C) $-1/2$ (D) -1

27. Let $f(x) = ||x| - 1|$, then points where $f(x)$ is differentiable is (are) [IIT - 2005]

- (A) $0, \pm 1$ (B) ± 1 (C) 0 (D) 1

28. If f is differentiable function satisfying

$f\left(\frac{1}{n}\right) = 0$ for all $n \geq 1, n \in \mathbb{I}$, then

[IIT - 2005]

- (A) $f(x) = f'(x) = 0, x \in (0, 1]$
(B) $f'(0) = 0 = f(0)$

(C) $f(0) = 0$ but $f'(0)$ not necessarily zero

(D) $|f(x)| \leq 1, x \in (0, 1]$

29. If $y = f(x)$ and $y \cos x + x \cos y = \pi$ for $x \in (-r, r)$, where $r > 0$, then the value of $f''(0)$ is [IIT - 2005]

- (A) π (B) $-\pi$ (C) 0 (D) 2π

30. $f''(x) = -f(x)$ where $f(x)$ is a continuous double differentiable function &

$g(x) = f'(x)$. If $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$

and $F(5) = 5$, then $F(10)$ is [IIT - 2006]

- (A) 0 (B) 5 (C) 10 (D) 25

31. If f is a periodic function with period T & $[.]$ denotes G.I.F, then,

$$\frac{L}{n^3} \left[f(x+T)^{f(x+T)} + 2^2 f(x+2T)^{f(x+2T)} + 3^2 f(x+3T)^{f(x+3T)} + \dots + n^2 f(x+nT)^{f(x+nT)} \right]$$

- A) $\frac{(f(x))^{f(x)}}{2}$ B) $\frac{(f(x))^{f(x)}}{4}$
C) $\frac{2(f(x))^{f(x)}}{3}$ D) $\frac{(f(x))^{f(x)}}{3}$

32. Let $g(x) = \frac{(x-1)^n}{\log \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0, n > 0$, and let p be the left hand derivative of $|x-1|$ at $x = 1$. If

$\lim_{x \rightarrow 1+} g(x) = p$, then [IIT-2008]

- (A) $n = 1, m = 1$ (B) $n = 1, m = -1$
(C) $n = 2, m = 2$ (D) $n > 2, m = n$

33. Number of points, where the function

$f(x) = \text{Max} \left\{ \text{sgn}(x), -\sqrt{(9-x^2)}, x^3 \right\}$ is

continuous but not differentiable is:

- (A) 6 (B) 5 (C) 4 (D) 3

34. The value of K such that the function

$f(x) = |x^2 + (k-1)|x| - k|$ is non-differentiable at exactly five points is

- A) 3 B) 4 C) -2 D) 5

35. A function $f: R \rightarrow R$ is defined as

$$f(x) = \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}, \text{ where } f \text{ is}$$

continuous on R . the values of a, b and c are

- A) $c = 1, a, b \in R$ B) $c = 0, a, b \in R$
C) $a = 0, b, c \in R$ D) $c = 3, a, b \in R$

36. A discontinuous function $y = f(x)$ satisfying $x^2 + y^2 = 4$ is given by

$$(A) f(x) = \begin{cases} \sqrt{4-x^2} & -2 \leq x < 0 \\ -\sqrt{4-x^2} & 0 \leq x \leq 2 \end{cases}$$

$$(B) f(x) = \begin{cases} -\sqrt{4+x^2} & -2 \leq x < 0 \\ \sqrt{4-x^2} & 0 \leq x \leq 2 \end{cases}$$

$$(C) f(x) = \sqrt{4-x^2} \quad -2 \leq x \leq 2$$

$$(D) f(x) = -\sqrt{4-x^2} \quad -2 \leq x \leq 2$$

37. If $f(x) = \begin{cases} x[x], & 0 \leq x < 2 \\ (x-1)[x], & 2 \leq x \leq 3 \end{cases}$, where $[.]$

denotes the greatest integer function, then

(A) both $f'(1)$ and $f'(2)$ do not exist

(B) $f'(1)$ exists but $f'(2)$ does not exist

(C) $f'(2)$ exists but $f'(1)$ does not exist

(D) both $f'(1)$ and $f'(2)$ exists

38. Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{x} \right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then f is

[IIT-2012]

- (A) differentiable both at $x = 0$ and at $x = 2$
(B) differentiable at $x = 0$ but not differentiable at $x = 2$

LIMITS, CONTINUITY & DIFFERENTIABILITY

(C) not differentiable at $x = 0$ differentiable at $x = 2$

(D) differentiable neither at $x = 0$ nor at $x = 2$

39. The function

$$f(x) = \begin{cases} \sin \left(\frac{\pi x}{2} \right), & x < 1 \\ |2x - 3| [x], & x \geq 1 \end{cases}$$

(A) is continuous at $x = 1$

(B) is differentiable at $x = 1$

(C) is continuous but not differentiable at $x = 1$

(D) $\lim_{x \rightarrow 1} f(x)$ does not exist

MULTIPLE ANSWER QUESTIONS

40. Let $f(x) = \frac{\sin^{-1}(1-\{x\}) \cdot \cos^{-1}(1-\{x\})}{\sqrt{2}\{x\} \cdot (1-\{x\})}$,

where $\{x\}$ denotes the fractional part of x . Then

$$(A) \lim_{x \rightarrow 0^+} f(x) = \infty \quad (B) \lim_{x \rightarrow 0^-} f(x) = \frac{\pi}{2\sqrt{2}}$$

$$(C) \lim_{x \rightarrow 0^+} f(x) = -\frac{\pi}{2} \quad (D) \lim_{x \rightarrow 0^-} f(x) = 0$$

41. If $\lim_{x \rightarrow 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 \log_e(1+x) - 2x^3 + x^4}$ exists and is finite then

- (A) $a = 6$ (B) $b = 6$
(C) $a(b+c) = 36$ (D) $c = 2$

42. Let $f(x) = [x]$ and $g(x) = \begin{cases} 0, & x \in I \\ x^2, & x \notin I \end{cases}$, then

(A) $\lim_{x \rightarrow 1} g(x)$ exists, but $g(x)$ is not continuous at $x = 1$

(B) $\lim_{x \rightarrow 1} f(x)$ does not exist and $f(x)$ is not continuous at $x = 1$

(C) $g \circ f$ is continuous for all x

(D) $f \circ g$ is continuous for all x

43. Which of the following function(s) has/have removable discontinuity at $x = 1$?

$$(A) f(x) = \frac{1}{\ln|x|} \quad (B) f(x) = \frac{x^2 - 1}{x^3 - 1}$$

$$(C) f(x) = 2^{-2^{1-x}} \quad (D) f(x) = \frac{\sqrt{x+1} - \sqrt{2x}}{x^2 - x}$$

44. The function $f(x) = ||2x - 3| - 10|$ is non differentiable at

$$(A) x \in \left\{ \frac{-7}{2}, \frac{13}{2} \right\} \quad (B) x \in \left\{ \frac{-7}{2}, \frac{13}{2}, \frac{3}{2} \right\}$$

$$(C) x \in \left\{ \frac{3}{2} \right\} \quad (D) x \in \left\{ -\frac{3}{2} \right\}$$

45. $f(x)$ is differentiable function and $(f(x), g(x))$ is differentiable at $x = a$, then

- (A) $g(x)$ must be differentiable at $x = a$
 (B) if $g(x)$ is discontinuous, then $f(a) = 0$
 (C) $f(a) \neq 0$, then $g(x)$ must be differentiable
 (D) nothing can be said

46. If $f(x) = \begin{cases} |x| - 3, & x < 1 \\ |x - 2| + a, & x \geq 1 \end{cases}$ and

$$g(x) = \begin{cases} 2 - |x|, & x < 2 \\ \operatorname{sgn}(x) - b, & x \geq 2 \end{cases} \text{ and}$$

$h(x) = f(x) + g(x)$ is discontinuous at exactly one point, then which of the following values of a and b are possible:

- (A) $a = -3, b = 0$ (B) $a = 2, b = 1$
 (C) $a = 2, b = 0$ (D) $a = -3, b = 1$

47. Let $[x]$ denote the greatest integer less than or equal to x . If $f(x) = [x \sin \pi x]$, then $f(x)$ is

- (A) continuous at $x = 0$
 (B) continuous in $(-1, 0)$
 (C) differentiable at $x = 1$
 (D) differentiable in $(-1, 1)$

48. If $f(x) = \min \{1, x^2, x^3\}$, then

- (A) $f(x)$ is continuous everywhere
 (B) $f(x)$ is continuous and differentiable everywhere
 (C) $f(x)$ is not differentiable at two points
 (D) $f(x)$ is not differentiable at one point

49. For a function

$$f(x) = \frac{\ln(\{\sin x\}\{\cos x\} + 1)}{\{\sin x\}\{\cos x\}}, \text{ where } \{ \cdot \}$$

denotes fractional part function, then

$$(A) f(0^-) = f\left(\frac{\pi^+}{2}\right) \quad (B) f(0^+) = f\left(\frac{\pi^-}{2}\right)$$

$$(C) \lim_{x \rightarrow 0} f(x) = 1 \quad (D) \lim_{x \rightarrow \pi/2} f(x) = 1$$

50. If $f(x) = \begin{cases} x \log \cos x & \text{for } x \neq 0 \\ \log(1 + x^2) & \text{for } x = 0 \end{cases}$, then

- (A) $f(x)$ is continuous at $x = 0$
 (B) $f(x)$ is continuous but not differentiable at $x = 0$
 (C) $f(x)$ is differentiable at $x = 0$
 (D) $f(x)$ is not continuous at $x = 0$

51. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$ ($[\cdot]$ denotes the greatest integer function) and $f(x)$ is non-constant continuous function, then

- (A) $\lim_{x \rightarrow a} f(x)$ is an integer
 (B) $\lim_{x \rightarrow a} f(x)$ is non-integer
 (C) $f(x)$ has local maximum at $x = a$
 (D) $f(x)$ has local minimum at $x = a$

52. Let $f(x) = \frac{1}{[\sin x]}$, (where $[\cdot]$ denotes the

- (A) domain of $f(x)$ is $(2n\pi + \pi, 2n\pi + 2\pi) \cup \{2n\pi + \pi/2\}$, where $n \in I$

- (B) $f(x)$ is continuous, when

$$x \in (2n\pi + \pi, 2n\pi + 2\pi), \text{ where } n \in I$$

- (C) $f(x)$ is differentiable at $x = \pi/2$
 (D) none of these

53. Let $f(x) = (x - \lambda)^m |x - \lambda|$, where m is a non negative integer, then at $x = \lambda$

- (A) $f(x)$ is not differentiable, if $m = 0$
 (B) $f(x)$ is differentiable, if $m \geq 1$
 (C) $f'(x)$ is not differentiable, if $m = 1$
 (D) $f'(x)$ is differentiable, if $m \geq 2$

54. If $f(x) = \begin{cases} x^2(\operatorname{sgn}[x] + \{x\}), & 0 \leq x < 2 \\ \sin x + |x - 3|, & 2 \leq x < 4 \end{cases}$
where $[]$ and $\{ \}$ represents the greatest integer and the fractional part function, respectively
- (A) $f(x)$ is differentiable at $x = 1$
 (B) $f(x)$ is continuous but non differentiable at $x = 1$
 (C) $f(x)$ is non differentiable at $x = 2$
 (D) $f(x)$ is discontinuous at $x = 2$

55. Consider the function $f(x) = \left(\frac{ax+1}{bx+2} \right)^x$,

where $a^2 + b^2 \neq 0$ then $\lim_{x \rightarrow \infty} f(x)$:

- (A) exists for all values of a and b
 (B) is zero for $0 < a < b$
 (C) is non-existent for $a > b > 0$
 (D) is $e^{-(1/a)}$ or $e^{-(1/b)}$ for $a = b$
56. A function $f(x)$ satisfies the relation $f(x+y) = f(x) + f(y) + xy(x+y), \forall x, y \in \mathbb{R}$
 If $f'(0) = 1$ then
- (A) $f(x)$ is a polynomial function
 (B) $f(x)$ is an exponential function
 (C) $f(x)$ is twice differentiable for all $x \in \mathbb{R}$
 (D) $f'(3) = 8$

57. If $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \leq x \leq 2 \\ 37 - x, & 2 < x \leq 3 \end{cases}$, then

- (A) $f(x)$ is increasing on $[-1, 2]$
 (B) $f(x)$ is continuous on $[-1, 3]$
 (C) $f'(2)$ does not exist
 (D) $f(x)$ has the maximum value at $x = 2$
58. Let $f(x+y) = f(x) + f(y) + 2xy - 1 \quad \forall x, y \in \mathbb{R}$. If $f(x)$ is differentiable and $f'(0) = \sin \phi$, then

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(A) $f(x) < 0 \quad \forall x \in \mathbb{R}$ (B) $f(x) > 0 \quad \forall x \in \mathbb{R}$

(C) $f(x) \geq \frac{3}{4} \quad \forall x \in \mathbb{R}$

(D) $-1 \leq f(x) \leq 1 \quad \forall x \in \mathbb{R}$

59. The function, $f(x) =$

$\max \{ (1-x), (1+x), 2 \}, x \in (-\infty, \infty)$ is

[IIT - 1995]

- (A) continuous at all points
 (B) differentiable at all points
 (C) differentiable at all points except at $x = 1$ and $x = -1$
 (D) continuous at all points except at $x = 1$ and $x = -1$, where it is discontinuous

60. Evaluate $\lim_{x \rightarrow 0} \left(\frac{(1 + \{x\})^{1/x}}{e} \right)^{1/\{x\}}$ if it

exists, where $\{x\}$ denotes the fractional part of x .

- (A) $2/e$ (B) $e^{-1/2}$
 (C) 2 (D) does not exist

61. Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then [IIT - 2009]

- (A) $a = 2$ (B) $a = 1$
 (C) $L = 1/64$ (D) $L = 1/34$

62. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y), \forall x, y \in \mathbb{R}$. If $f(x)$ is differentiable at $x = 0$, then [IIT-2011]

- (A) $f(x)$ is differentiable only in a finite interval containing zero
 (B) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (C) $f'(x)$ is continuous $\forall x \in \mathbb{R}$
 (D) $f(x)$ is differentiable except at finitely many points

63. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0 \\ x-1, & 0 < x \leq 1 \\ \ln x, & x > 1 \end{cases}$, then [IIT-2011]

- (A) $f(x)$ is continuous at $x = -\frac{\pi}{2}$
 (B) $f(x)$ is not differentiable at $x = 0$
 (C) $f(x)$ is differentiable at $x = 1$
 (D) $f(x)$ is differentiable at $x = -3/2$

64. For every integer n , let a_n and b_n be real numbers. Let function $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases},$$

for all integers n . If f is continuous, then which of the following hold(s) for all n [IIT-2012]

- (A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$
 (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

COMPREHENSIONS QUESTIONS

Passage - 1:

Let $f(x) = \lim_{n \rightarrow \infty} \left(\cos \sqrt{\frac{x}{n}} \right)^n$,

$g(x) = \lim_{n \rightarrow \infty} (1 + x + x^n \sqrt[n]{e})^n$. Now, consider

the function $y = h(x)$, where

$$h(x) = \tan^{-1} (g^{-1}(f^{-1}(x))).$$

65. $\lim_{x \rightarrow 0} \frac{\ln(f(x))}{\ln(g(x))}$ is equal to

- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 0 (D) 1

66. Domain of the function $y = h(x)$ is
 (A) $(0, \infty)$ (B) $\mathbb{R}$ (C) $(0, 1)$ (D) $[0, 1]$

67. Range of the function $y = h(x)$ is
 (A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2}, 0\right)$
 (C) $\mathbb{R}$ (D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Passage - 2:

Let $f(x) = \begin{cases} x+2, & 0 \leq x < 2 \\ 6-x, & x \geq 2 \end{cases}$,

$$g(x) = \begin{cases} 1 + \tan x, & 0 \leq x < \frac{\pi}{4} \\ 3 - \cot x, & \frac{\pi}{4} \leq x < \pi \end{cases}$$

68. $f(g(x))$ is

- (A) discontinuous at $x = \frac{\pi}{4}$
 (B) differentiable at $x = \frac{\pi}{4}$
 (C) continuous but non differentiable at $x = \frac{\pi}{4}$
 (D) differentiable at $x = \frac{\pi}{4}$, but derivative is not continuous.

69. The number of points of non differentiability of $h(x) = |f(g(x))|$ is
 (A) 1 (B) 2 (C) 3 (D) 4

70. The range of $h(x) = f(g(x))$ is
 (A) $(-\infty, \infty)$ (B) $(4, \infty)$
 (C) $(-\infty, 4]$ (D) $[4, \infty)$

Passage - 3:

Let $f(x) = \frac{\sin^{-1}(1-\{x\}) \cdot \cos^{-1}(1-\{x\})}{\sqrt{2}\{x\} \cdot (1-\{x\})}$,

where $\{.\}$ denotes the fractional part function.

71. If $R = \lim_{x \rightarrow 0^+} f(x)$, then the value of $\cos(100R)$ is :
A) -1 B) 0 C) 1/2 D) 1
72. If $L = \lim_{x \rightarrow 0^-} f(x)$, then the value of $\sin(99\sqrt{2}L)$ is :
A) -1 B) 0 C) 1/2 D) 1
73. The value of $[2R^2 + 4L^2]$ is [where $[.]$ denotes the greatest integer function] :
A) 3 B) 6 C) 9 D) 12

Passage - 4:

$$\text{Let } f(x) = \begin{cases} [x], & -2 \leq x \leq -\frac{1}{2} \\ 2x^2 - 1, & -\frac{1}{2} < x \leq 2 \end{cases} \text{ and}$$

$g(x) = f(|x|) + |f(x)|$, where $[.]$ represents greatest integer function.

74. The number of points where $|f(x)|$ is non-differentiable is
A) 3 B) 4 C) 2 D) 5
75. The number of points where $g(x)$ is non-differentiable is
A) 4 B) 5 C) 2 D) 3
76. The number of points where $g(x)$ is discontinuous is
A) 1 B) 2 C) 3 D) 0

MATRIX-MATCH QUESTIONS

The statements in Column I are labelled A, B, C and D, while the statements in Column II are labelled p, q, r, s and t. Any given statement in Column I can have correct matching with ONE OR MORE statements(s) in Column II. The appropriate bubbles corresponding to the answers to these equations have to be darkened as illustrated in the following example. If the correct matches are A - p, s and t; B - q and

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r; C - p and q; and D - s and t; then the correct darkening of bubbles will look like the following.

	p	q	r	s	t
A	☒	☒	☐	☒	☒
B	☐	☒	☐	☐	☐
C	☐	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

77. Column - I

- (A) If $\lim_{x \rightarrow \infty} (\sqrt{x^2 - x - 1} - ax - b) = 0$, where $a > 0$, then there exists at least one a and b for which point $(a, 2b)$ lies on the line
- (B) If $\lim_{x \rightarrow \infty} \frac{(1+a^3) + 8e^{1/x}}{1 + (1-b^3)e^{1/x}} = 2$, then there exists at least one a and b for which point (a, b^3) lies on the line
- (C) If $\lim_{x \rightarrow \infty} (\sqrt{x^4 - x^2 + 1} - ax^2 - b) = 0$, then there exist at least one a and b for which point $(a, -2b)$ lies on the line
- (D) If $\lim_{x \rightarrow -a} \frac{x^7 + a^7}{x + a} = 7$, where $a < 0$, then there exists at least one a for which point $(a, 2)$ lies on the line.

Column - II

- (P) $y = -3$
(Q) $3x - 2y - 5 = 0$
(R) $15x - 2y - 11 = 0$
(S) $y = 2$

78. Column - I

- (A) $\lim_{x \rightarrow \frac{\pi^+}{2}} \tan^{-1}(\tan x)$
- (B) $\lim_{n \rightarrow \infty} \left[\sum_{r=1}^n \frac{1}{2^r} \right]$ ($[.]$ denotes the greatest integer function)
- (C) $\lim_{x \rightarrow \infty} \sec^{-1} \left(\frac{x}{x+1} \right)$
- (D) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{(1 - \sin x)^{2/3}}$

Column - II

(P) 0

(Q) does not exist

(R) $-\frac{\pi}{2}$

(S) $\frac{\pi}{2}$

79. Column - I

(A) Number of points where the function

$$f(x) = \begin{cases} 1 + \left[\cos \frac{\pi x}{2} \right] & 1 < x \leq 2 \\ 1 - \{x\} & 0 \leq x < 1 \\ |\sin \pi x| & -1 \leq x < 0 \end{cases}$$

([.] denotes the greatest integer function)

{ } denotes fractional part of x

and $f(1) = 0$ is continuous but non differentiable

$$(B) f(x) = \begin{cases} x^2 e^{1/x}, & x \neq 0 \\ 0, & x = 0 \end{cases}, \text{ then } f(0^-) =$$

(C) The number of points at which

$$g(x) = \frac{1}{1 + \frac{2}{f(x)}} \text{ is not differentiable}$$

$$\text{where } f(x) = \frac{1}{1 + \frac{1}{x}} \text{ is}$$

(D) Number of points where tangent

does not exist for the curve $y = \operatorname{sgn}(x^2 - 1)$ **Column - II**

(P) 0

(Q) 1

(R) 2

(S) 3

80. Column - I

$$(A) f(x) = \lim_{x \rightarrow 1/2} \left\{ \lim_{n \rightarrow \infty} \cos^{2n}(2\pi x) + \left\{ x + \frac{1}{2} \right\} \right\},$$

where { } denotes the fractional part

$$(B) f(x) = (\log_e x)(x-1)^{1/5} \text{ at } x = 1$$

$$(C) f(x) = [\cos 2\pi x] + \sqrt{\left\{ \sin \pi \frac{x}{2} \right\}},$$

where [.] and { } denote the greatest integer and the fractional part function respectively at $x = 1$

$$(D) f(x) = \begin{cases} \cos 2x, & x \in \mathbb{Q} \\ \sin x, & x \notin \mathbb{Q} \end{cases} \text{ at } x = \frac{\pi}{6}$$

Column - II

(P) continuous

(Q) discontinuous

(R) differentiable

(S) non differentiable

81. Column I (functions)

$$(A) f(x) = |x|$$

$$(B) f(x) = x^n |x|, n \in \mathbb{N}$$

$$(C) f(x) = \begin{cases} x \ln |\sin x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$(D) f(x) = \begin{cases} x e^{1/x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Column II (properties)(P) continuous at $x = 0$ (Q) Discontinuous at $x = 0$ (R) differentiable at $x = 0$ (S) non-differentiable at $x = 0$ **82. Column - I**

$$(A) \text{ Let } f(x) = \begin{cases} x^6, & x > 1 \\ x^3, & x \leq 1 \end{cases} \text{ Then } f(x) \text{ is}$$

differentiable at $x =$

(B) Number of points of non-differentiability of $f(x) = \min\{2, x^2, x^3\}$ is

(C) $f(x) = x^2 \sin\left(\frac{1}{x}\right)$, $x \neq 0$, $f(0) = 0$ then

$f'(0^-)$ is

(D) $f(x) = |x-1| + |x| + |x+1|$, then $f'(0^+)$ is

Column-II

(P) 0

(Q) 1

(R) 2

(S) 3

(T) 5

83. **Column I (functions)** [IIT-2007]

(A) $x|x|$

(B) $\sqrt{|x|}$

(C) $x + [x]$, $[.] \rightarrow \text{G.I.F.}$

(D) $|x-1| + |x+1|$

Column II (properties)

(P) continuous in $(-1, 1)$

(Q) differentiable in $(-1, 1)$

(R) strictly increasing in $(-1, 1)$

(S) not differentiable at least at one point in $(-1, 1)$

FILL IN BLANKS QUESTIONS

84. Let $f(x) = x|x|$. The set of points where $f(x)$ is twice differentiable is [IIT - 92]

85. Let $f(x) = (x-1)(x-2)(x-3)\dots(x-n)$, $n \in \mathbb{N}$ and $f'(n) = 5040$ then 'n' is

86. Let $f(xy) = xf(y) + yf(x)$ for all $x, y \in \mathbb{R}_+$ and $f(x)$ be differentiable in $(0, \infty)$, then determine $f(x)$.

ASSERTION-REASONING QUESTIONS

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for

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Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False

(D) Statement -1 is False, statement-2 is True.

87. Statements-1: The set of all points where

$$\text{the function } f(x) = \begin{cases} 0, & x = 0 \\ \frac{x}{1+e^{1/x}}, & x \neq 0 \end{cases} \text{ is}$$

differentiable is $(-\infty, \infty)$.

Statements-2: $Lf'(0) = 1$, $Rf'(0) = 0$ and

$$f'(x) = \frac{1+e^{1/x} + e^{1/x} \frac{1}{x}}{(1+e^{1/x})^2}, \text{ which exists when } x \neq 0.$$

88. Statements-1: $f(x) = \begin{cases} 3-x^2, & x > 1 \\ x^3+1, & x \leq 1 \end{cases}$ then $f(x)$

is differentiable at $x = 1$

Statements-2: A function $y = f(x)$ is said to have a derivative if

$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

89. Statement - 1: $\lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$ does not exist

Statement - 2: L.H.L. = 1 and R.H.L. = -1.

90. Consider the function $f(x) = (|x| - |x-1|)^2$ Statement - 1: $f(x)$ is continuous everywhere but not differentiable at $x = 0$ and 1.

Statement - 2: $f'(0^-) = 0$, $f'(0^+) = -4$, $f'(1^-) = 4$, $f'(1^+) = 0$.

91. Let f and g be real valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$, and $f(x) = g(x) \sin x$.

Statment-1:

$$\lim_{x \rightarrow 0} [g(x) \cot x - g(x) \operatorname{cosec} x] = f''(0). \text{ and}$$

Statment-2: $f'(0) = g(0)$

92. Statement 1 : If $f(x)$ and $g(x)$ are continuous and differentiable function, then

there exist $c, c \in [a, b]$ such that

$$\frac{f'(c)}{f(a)-f(c)} + \frac{g'(c)}{g(b)-g(c)} = 1$$

Statement 2: If $f(x)$ and $g(x)$ are continuous and differentiable function, then $h(x) = (f(a) - f(x))(g(b) - g(x))e^x$ is continuous and differentiable.

93. **Statement 1 :** The function

$$f(x) = (3x-1)|4x^2 - 12x + 5| \cos \pi x \text{ is}$$

differentiable at $x = \frac{1}{2}, \frac{5}{2}$.

Statement 2: $\cos(2n+1)\frac{\pi}{2} = 0 \forall n \in \mathbb{I}$

94. **Statement 1 :** $\lim_{x \rightarrow a} f(x)$ exists = k , but

$\lim_{x \rightarrow k} g(x)$ does not exist. If $\lim_{x \rightarrow a} g(f(x))$

exists, then $x = a$ is a point of extremum for $y = f(x)$, If $f(x)$ is non linear

Statement 2: $\lim_{x \rightarrow k} g(x)$ does not exist but

$\lim_{x \rightarrow a} g(f(x))$ exists, $f(x)$ will approach k

when $x \rightarrow a$ through only one side.

95. Let $f(0) = 0, f\left(\frac{\pi}{2}\right) = 1, f\left(\frac{3\pi}{2}\right) = -1$ be a continuous and twice differentiable function.

Statement 1 : $|f''(x)| \leq 1$ for at least one

$$x \in \left(0, \frac{3\pi}{2}\right)$$

Statement 2: According to Rolle's theorem

if $y = g(x)$ is continuous and differentiable $\forall x \in [a, b]$ and $g(a) = g(b)$, then there exist at least one c such that $g'(c) = 0$.

INTEGER QUESTIONS

This section contains 9 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following.

X	Y	Z	W
☐ 0	☒ 0	☐ 0	☐ 0
☐ 1	☐ 1	☐ 1	☐ 1
☐ 2	☐ 2	☒ 2	☐ 2
☐ 3	☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5	☐ 5
☒ 6	☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☒ 9	☐ 9

96. Let $f(x+y) = f(x) + f(y) - 2xy - 1$ for all x and y . if $f'(0)$ exists and $f'(0) = -\sin \alpha$, then the value of $f\{f'(0)\}$ is

97. Let $f(x)$ be a function satisfying the condition $f(-x) = f(x)$ for all real x . If $f'(0)$ exists, its value is

98. Let ' f ' be a continuous function on $\mathbb{R}$.

If $f(1/4^n) = (\sin e^n)e^{-n^2} + \frac{n^2}{n^2+1}$, then $f(0)$ is

99. If $f(x)$ be a twice differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that $t^2 f(x) - 2tf'(x) + f''(x) = 0$ has two equal values of t for all x and $f(0) = 1, f'(0) = 2$,

then $\left(\lim_{x \rightarrow 0} \frac{f(x)-1}{x} - \frac{t}{2}\right)$ is

100. Let $L =$

$$\lim_{x \rightarrow 0} \frac{\sin x + \ln(\sqrt{1 + \sin^2 x}) - \sin x}{\sin^3 x} \text{ then the}$$

value of $6L + 5$ is

101. The minimum area bounded by the function $y = f(x)$ and $y = ax + 9$ ($a \in \mathbb{R}$) where f satisfies the relation

$$f(x + y) = 2f(x) + xf(y) + y\sqrt{f(x)} \quad \forall x, y \in \mathbb{R}$$

and $f'(0) = 0$ is $9A$, then the value of A is

102. Let $f(x) = x + \cos x + 2$ and $g(x)$ be the inverse function of $f(x)$ then $g'(3)$ equals

103. $\frac{f(x+2y)}{3} = \frac{f(x) + 2f(y)}{3} \quad \forall x, y \in \mathbb{R}.$

If $f'(0) = 1$, $f(0) = 2$, then $f(2)$ is

104. The number of points of discontinuity of $f(x) = [2 \cos x]$, $x \in (0, 2\pi)$,

($[.]$ represents the greatest integer function) are (where $[.]$ represents greatest integer function)

105. The number of points of discontinuity of the function $f(x) = \left[\frac{6x}{\pi}\right] \cos\left[\frac{3x}{\pi}\right]$, in the interval

$$\left(\frac{\pi}{10}, \frac{11\pi}{10}\right) \text{ is} \quad [\text{IIT - 2009}]$$

($[.]$ represents the greatest integer function)

106. Let $p(x)$ be a polynomial of degree 4 having

$$\text{extremum at } x = 1, 2 \text{ and } \lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2}\right) = 2.$$

Then the value of $p(2)$ is [IIT - 2009]

107. If the function $f(x) = x^3 + e^{x/2}$ and

$$g(x) = f^{-1}(x), \text{ then } g'(1) \text{ is} \quad [\text{IIT - 2009}]$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

- | | | | |
|---------|---------|---------|---------|
| 1. (B) | 2. (A) | 3. (C) | 4. (C) |
| 5. (D) | 6. (A) | 7. (C) | 8. (A) |
| 9. (A) | 10. (D) | 11. (B) | 12. (B) |
| 13. (D) | 14. (A) | 15. (B) | 16. (B) |
| 17. (A) | 18. (B) | 19. (C) | 20. (D) |
| 21. (D) | 22. (A) | 23. (C) | 24. (B) |
| 25. (D) | 26. (D) | 27. (A) | 28. (B) |
| 29. (A) | 30. (B) | 31. (D) | 32. (C) |
| 33. (B) | 34. (C) | 35. (A) | 36. (A) |
| 37. (A) | 38. (B) | 39. (C) | |

MULTIPLE ANSWER QUESTIONS

- | | |
|---------------|------------------|
| 40. (A, B) | 41. (A, B, C) |
| 42. (A, B, C) | 43. (B, D) |
| 44. (A, B, C) | 45. (B, C) |
| 46. (A, B, C) | 47. (A, B, D) |
| 48. (A, D) | 49. (A, B) |
| 50. (A, C) | 51. (A, D) |
| 52. (A, B) | 53. (A, B, C, D) |
| 54. (A, C, D) | 55. (B, C, D) |
| 56. (A, C, D) | 57. (A, B, D) |
| 58. (B, C) | 59. (A, C) |
| 60. D | 61. (A, C) |
| 62. (B, C) | 63. (A, B, C, D) |
| 64. (B, D) | |

COMPREHENSIONS QUESTIONS

- | | | |
|---------|---------|---------|
| 65. (B) | 66. (C) | 67. (D) |
| 68. (C) | 69. (B) | 70. (C) |
| 71. (D) | 72. (A) | 73. (C) |
| 74. (C) | 75. (D) | 76. (B) |

MATRIX-MATCH QUESTIONS

77. A-q, B-p, C-q, D-s
 78. A-s, B-p, C-p, D-q
 79. A-q, B-r, C-p, D-q
 80. (A - q,s), (B - p,r), (C - p, r), (D - p,s)
 81. (A - p,s), (B - p,r), (C - p,s), (D - q,s)
 82. A-p, B-r, C-p, D-q
 83. (A - p,q,r), (B - p,s), (C - r,s), (D - p,q)

FILL IN BLANKS QUESTIONS

84. $\mathbb{R} - \{0\}$ 85. 8
 86. $f(x) = f'(1)(x \ln x)$

KEY - LEVEL - V SINGLE ANSWER QUESTIONS

ASSERTION-REASONING QUESTIONS

87. (D) 88. (D) 89. (C) 90. (A)
 91. (B) 92. (D) 93. (A) 94. (D)
 95. (A)

INTEGER QUESTIONS

96. 1 97. 0 98. 1 99. 1
 100. 6 101. 8 102. 1 103. 4
 104. 6 105. 6 106. 0 107. 2

HINTS

LEVEL-V

SINGLE ANSWER QUESTIONS

1. Given $\lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x - e^x \cos x + e^x - \frac{x^3}{2}}{x^n}$

$$= \lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x) - \frac{x^3}{2}}{x^n}$$

$$= \lim_{x \rightarrow 0} \frac{\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots - 1\right)}{x^n}$$

$$\times \frac{\left[\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) - \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} - \dots\right)\right] - \frac{x^3}{2}}{x^n}$$

$$\lim_{x \rightarrow 0} \frac{\left(\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \dots\right) \left[-x - x^2 - \frac{x^3}{3!} - \frac{2x^5}{5!} - \dots\right] - \frac{x^3}{2}}{x^n}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{x^3}{2} + \frac{x^4}{2} + \frac{x^5}{12} - \frac{x^5}{24} + 0 \dots\right) - \frac{x^3}{2}}{x^n}$$

= non zero if $n = 4$

2. Given

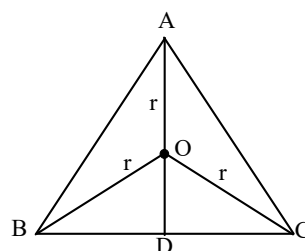
$$\lim_{x \rightarrow \infty} \frac{2x^{1/2} + 3x^{1/3} + \dots + nx^{1/n}}{(2x-3)^{1/2} + (2x-3)^{1/3} + \dots + (2x-3)^{1/n}}$$

$$= \lim_{h \rightarrow 0} \frac{2\left(\frac{1}{h^{1/2}}\right) + 3\left(\frac{1}{h^{1/3}}\right) + \dots + n\left(\frac{1}{h^{1/n}}\right)}{\frac{1}{h^{1/2}}(2-3h)^{1/2} + \frac{1}{h^{1/3}}(2-3h)^{1/3} + \dots + \frac{1}{h^{1/n}}(2-3h)^{1/n}}$$

[on putting $x = \frac{1}{h}$ as $x \rightarrow \infty, h \rightarrow 0$]

$$= \lim_{h \rightarrow 0} \frac{2 + 3h^{\left(\frac{1}{2} - \frac{1}{3}\right)} + 4h^{\left(\frac{1}{2} - \frac{1}{4}\right)} + \dots + nh^{\left(\frac{1}{2} - \frac{1}{n}\right)}}{(2-3h)^{1/2} + h^{\left(\frac{1}{2} - \frac{1}{3}\right)}(2-3h)^{1/3} + \dots + h^{\left(\frac{1}{2} - \frac{1}{n}\right)}(2-3h)^{1/n}}$$

$$= \frac{2 + 0 + 0 + 0 + \dots}{2^{1/2} + 0 + 0 + 0 + \dots} = \sqrt{2}$$



3.

In $\triangle ABC$, $AB = AC$

$AD \perp BC$ (D is mid point of BC)

Let r = radius of circumcircle

$\Rightarrow OA = OB = OC = r$

$$\text{Now } BD = \sqrt{BO^2 - OD^2} = \sqrt{r^2 - (h-r)^2} = \sqrt{2r-h^2} \Rightarrow BC = 2\sqrt{2rh-h^2}$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2} \times BC \times AD = h\sqrt{2rh-h^2}$$

$$\text{Also } \lim_{h \rightarrow 0} \frac{A}{p^3} = \frac{h\sqrt{2rh-h^2}}{8(\sqrt{2rh-h^2} + \sqrt{2hr})^3}$$

$$= \lim_{h \rightarrow 0} \frac{h^{3/2} \sqrt{2r-h}}{8h^{3/2} (\sqrt{2r-h} + \sqrt{2r})^3}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{2r-h}}{8[\sqrt{2r-h} + \sqrt{2r}]^3}$$

$$= \frac{\sqrt{2r}}{8(\sqrt{2r} + \sqrt{2r})^3} = \frac{\sqrt{2r}}{8 \cdot 8 \cdot 2r \cdot \sqrt{2r}} = \frac{1}{128r}$$

4. given that,

$$\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n} = \text{finite non zero}$$

number

$$= \lim_{x \rightarrow 0} \frac{(\cos x - 1)(1 + \cos x)(e^x - \cos x)}{x^n(1 + \cos x)}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} \right) \cdot \left(\frac{e^x - \cos x}{x^{n-2}} \right) \cdot \left(\frac{1}{1 + \cos x} \right) \\
&= \lim_{x \rightarrow 0} \frac{1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots}{x^{n-2}} \cdot \frac{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots}{x^{n-2}} \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\left(1 + x + \frac{x^2}{3!} + \frac{2x^3}{4!} + \dots \right)}{x^{n-3}}
\end{aligned}$$

for this limit to be finite $n - 3 = 0 \Rightarrow n = 3$

5. $A_i = \frac{x - a_i}{|x - a_i|}, i = 1, 2, 3, \dots, n$

$$a_1 < a_2 < a_3 < \dots < a_n.$$

If x is in the left neighbourhood of

$$a_1 < a_2 < \dots < a_{m-1} < x < a_m < a_{m+1} < \dots < a_n$$

$$A_i = \frac{x - a_i}{x - a_i} = 1, i = 1, 2, \dots, m - 1$$

$$A_i = \frac{x - a_i}{(a_i - x)} = -1, i = m, m + 1, \dots, n$$

$$\therefore A_1 A_2 \dots A_n = (-1)^{n-m+1}$$

If x is in the right neighbourhood of a_m

$$a_1 < a_2 < \dots < a_{m-1} < a_m < x < a_{m+1} < \dots < a_n$$

$$A_i = \frac{x - a_i}{x - a_i} = 1, i = 1, 2, \dots, n$$

$$\therefore A_1 A_2 \dots A_n = (-1)^{n-m}$$

$$\therefore \lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n) = (-1)^{n-m}$$

$$\therefore \text{LHL} \neq \text{RHL}$$

Hence, $\lim_{x \rightarrow a_m} (A_1 A_2 \dots A_n)$ does not exist.

6. Let $y = \frac{x}{x + \frac{\sqrt[3]{x}}{x + \sqrt[3]{x}} \dots \infty}$

$$= \frac{x}{x + \frac{1}{x^{2/3}} \times \frac{x}{x + \sqrt[3]{x}} \dots \infty} = \frac{x}{x + \frac{y}{x^{2/3}}}$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$\Rightarrow y = \frac{x^{5/3}}{x^{5/3} + y} \Rightarrow y^2 + (x^{5/3})y - x^{5/3} = 0.$$

$$\begin{aligned}
\therefore \Rightarrow y &= \frac{-x^{5/3} \pm \sqrt{x^{10/3} + 4x^{5/3}}}{2} \\
&= \frac{-x^{5/3} + \sqrt{x^{10/3} + 4x^{5/3}}}{2} (\because y > 0)
\end{aligned}$$

$$= \frac{4x^{5/3}}{2\sqrt{(x^{10/3} + 4x^{5/3}) + x^{5/3}}} = \frac{2}{\sqrt{\left(1 + \frac{4}{x^{5/3}}\right) + 1}}$$

$$\text{Hence, } \lim_{x \rightarrow \infty} \frac{2}{\sqrt{1 + 0 + 1}} = \frac{2}{2} = 1$$

7. $L = \lim_{x \rightarrow 0} = - \lim_{x \rightarrow 0} \frac{D \prod_{r=2}^n (\cos rx)^{1/r}}{2x}$

$$\text{Let } y = \prod_{r=2}^n (\cos rx)^{1/r}$$

$$\Rightarrow \ln y = \sum_{r=2}^n \left(\frac{1}{r} \ln (\cos rx) \right)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = - \sum_{r=2}^n \tan (rx)$$

$$\Rightarrow -Dy = y \sum_{r=2}^n \tan (rx)$$

$$\Rightarrow D \prod_{r=1}^n (\cos rx)^{1/r} = -y \sum_{r=2}^n \tan (rx)$$

$$\Rightarrow L = \lim_{x \rightarrow 0} \frac{y \sum_{r=2}^n \tan (rx)}{2x}$$

$$= \frac{1}{2} (2 + 3 + 4 + \dots + n)$$

$$= \frac{1}{2} \left[\frac{n(n+1)}{2} - 1 \right] = \frac{n^2 + n - 2}{4}$$

$$\text{Now } \frac{n^2 + n - 2}{4} = 10 \Rightarrow n^2 + n - 42 = 0$$

$$\Rightarrow (n+7)(n-6) = 0 \Rightarrow n = 6$$

8. Put $a_1 = \cot \frac{\pi}{2}$, we get $a_n = \cot \frac{\pi}{2^n}$

then $\lim_{n \rightarrow \infty} \frac{a_n}{2^{n-1}} = \frac{2}{\pi}$

9. For existence of limit, $(3+a)=0$, $a=-3$

given limit $= \lim_{x \rightarrow 0} \frac{\sin 3x - 3x + bx^3}{x^3}$, $b = \frac{9}{2}$

10. $e^{\ln(1+b^2)} = 2b \sin^2 \theta \Rightarrow \sin^2 \theta = \frac{1+b^2}{2b}$
 $\Rightarrow \sin^2 \theta = 1$ as $\frac{1+b^2}{2b} \geq 1$, $\theta = \pm \frac{\pi}{2}$

11. Given $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4$
 $\Rightarrow \lim_{x \rightarrow \infty} \frac{x^2 + x + 1 - ax^2 - ax - bx - b}{(x + 1)} = 4$
 $\Rightarrow \lim_{x \rightarrow \infty} \frac{(1-a)x^2 + (1-a-b)x + (1-b)}{(x + 1)} = 4$
 $\Rightarrow 1-a=0$ and $1-a-b=4 \Rightarrow b=-4$, $a=1$.

12. Let $1+a=y$
 $\Rightarrow (y^{1/3} - 1)x^2 + (y^{1/2} - 1)x + y^{1/6} - 1 = 0$
 $\Rightarrow \left(\frac{y^{1/3} - 1}{y - 1} \right)x^2 + \left(\frac{y^{1/2} - 1}{y - 1} \right)x + \frac{y^{1/6} - 1}{y - 1} = 0$

Now taking $\lim_{y \rightarrow 1}$ on both the sides

$$\Rightarrow \frac{1}{3}x^2 + \frac{1}{2}x + \frac{1}{6} = 0 \Rightarrow 2x^2 + 3x + 1 = 0$$

$$x = -1, -\frac{1}{2}.$$

13. At $x=1$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f(1-h) = \lim_{h \rightarrow 0} (1-h)[1-h] = 0$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} (1+h)[1+h] = 1$$

and $f(1)=1$

$\therefore f(x)$ is discontinuous function at $x=1$ obviously it is not differentiable at $x=1$

At $x=2$,

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f(2-h) = \lim_{h \rightarrow 0} (2-h)[2-h] = 2$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} (1+h)[2+h] = 2$$

$$\lim_{x \rightarrow 2} f(x) = 2, \therefore f(x) \text{ is continuous at } x=2$$

$$\text{L.H.D.} = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} = \lim_{h \rightarrow 0} \frac{(2-h) - 2}{-h} = 1$$

$$\text{R.H.D.} = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{(2+h-1)2-2}{h} = 2$$

$\therefore f(x)$ is not differentiable at $x=2$

14. $f'(0^-) = \lim_{h \rightarrow 0^+} \frac{f(0-h) - f(0)}{-h} = 1$

$$f'(0^+) = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = 0$$

15. By graph of $y = \sin x$, $y = \cos x$, $y = 3/4$, we get graph of $f(x)$ and then we get two points of non differentiability.

16. The given function is clearly continuous at all points except possibly at $x = \pm 1$. As $f(x)$ is an even function, so we need to check its continuity only at $x = 1$.

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

$$\Rightarrow \lim_{x \rightarrow 1^-} (ax^2 + b) = \lim_{x \rightarrow 1^+} \frac{1}{|x|}$$

$$\Rightarrow a + b = 1 \quad \dots (i)$$

Clearly, $f(x)$ is differentiable for all x , except possibly at $x = \pm 1$. As $f(x)$ is an even function, so we need to check its differentiability at $x = 1$ only.

$$\lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1^-} \frac{ax^2 + b - 1}{x - 1} = \lim_{x \rightarrow 1^+} \frac{\frac{1}{|x|} - 1}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{ax^2 - a}{x - 1} = \lim_{x \rightarrow 1} \frac{-1}{x} \Rightarrow 2a = -1$$

$$\Rightarrow a = -1/2$$

Putting $a = -1/2$ in (i), we get $b = 3/2$.

17. $3 \leq 3 + 2 \cos x \leq 5$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f(x) = [3 + 2 \cos x]$ is discontinuous at those points where $3 + 2 \cos x$ is an integer.

Now, $3+2\cos x = 3$, if $\cos x = 0$. So,

$$x = -\frac{\pi}{2}, \frac{\pi}{2} \text{ (Not possible)}$$

$$3+\cos x = 4, \text{ if } \cos x = \frac{1}{2}.$$

So, x have two values $\frac{\pi}{3}$ and $-\frac{\pi}{3}$

$3+2\cos x = 5$, if $\cos x = 1$. so, $x = 0$

The number of values of $x = 2+1 = 3$

Hence, (A) is correct.

18. We have $f(x + y^{2n+1}) = f(x) + \{f(y)\}^{2n+1}$

..(i)

$$\Rightarrow f(0) = f(0) + \{f(0)\}^{2n+1}$$

$$[\text{putting } x = y = 0] \Rightarrow f(0) = 0$$

Also, from Eq(i)

$$f(1) = f(0) + \{f(1)\}^{2n+1}$$

$$[\text{putting } x = 0, y = 1]$$

$$= 0 + \{f(1)\}^{2n+1} \quad [\because f(0) = 0]$$

$$\Rightarrow f(1) [\{f(1)\}^{2n} - 1] = 0$$

$$\Rightarrow f(1) = 0 \text{ or } 1 \text{ or } -1$$

$$\text{Now, } f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{f(x)}{x}$$

$$\text{Since, } f'(0) \geq 0 \Rightarrow \lim_{x \rightarrow 0} \frac{f(x)}{0} \geq 0$$

$$\Rightarrow f(x) \geq 0 \text{ for } x > 0.$$

$$\text{Also } f(1) = 0 \Rightarrow f(x+1) = f(x) + \{f(1)\}^{2n+1}$$

$$= f(x) = f(2) = f(3) = \dots = 0, \text{ which is not}$$

true. Thus, $f(1) = 1$.

$$\Rightarrow f(x+1) = f(x) + 1$$

$\Rightarrow f(x)$ is a function whose value is

increased by 1 when x is increased by 1,

It is a linear function whose graph is a straight

line passing through origin ($\because f(0) = 0$)

having slope 1.

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$\text{Hence } f(x) = x, \therefore f'(6) = 1$$

19. Taking $x = y = 1$, we get $f(1)f(1) - f(1) = 2$

$$\Rightarrow f^2(1) - f(1) - 2 = 0 \Rightarrow (f(1) - 2)(f(1) + 1) = 0$$

$$\Rightarrow f(1) = 2 \text{ (as } f(1) > 0)$$

Taking $y = 1$, we get

$$f(x).f(1) - f(x) = x + 1$$

$$\Rightarrow f(x) = x + 1 \Rightarrow f^{-1}(x) = x - 1$$

$$\therefore f(x).f^{-1}(x) = x^2 - 1$$

20. $f(x) = \frac{x^2 - 1}{x^2 + 1}$

$$\lim_{x \rightarrow 0} f(x) = -1 \text{ and } \lim_{x \rightarrow 0} f(x) = 1$$

$$\therefore f(x) \in [-1, 1]$$

$\therefore$ minimum value of $f = -1$.

21. $f(x) = [x]^2 - [x^2]$

Let $x = m, m \in I$

$$\lim_{x \rightarrow m^-} f(x) = (m-1)^2 - (m^2 - 1) = 2 - 2m$$

$$\lim_{x \rightarrow m^+} f(x) = (m^2 - m^2) = 0$$

$\therefore f$ is continuous only at $m=1$.

22. (i) $\text{LD} = \lim_{h \rightarrow 0} \frac{f(k) - f(k-h)}{h}$

($k = \text{integer}$)

$$= \lim_{h \rightarrow 0} \frac{[k] \sin k\pi - [k-h] \sin(k-h)\pi}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-(k-1) \sin(k-h)\pi}{h} \quad [\because \sin k\pi = 0]$$

$$= \lim_{h \rightarrow 0} \frac{-(k-1) \sin(k\pi - h\pi)}{h}$$

$$[\sin(k\pi - \theta) = (-1)^{k-1} \sin \theta]$$

$$= \lim_{h \rightarrow 0} \frac{-(k-1)(-1)^k \sinh \pi}{h\pi} \times \pi$$

$$= \pi(k-1)(-1)^{k-1}$$

$$(ii) f(x) = \cos |x| + |x| = \begin{cases} \cos x - x, & x < 0 \\ \cos x + x, & x \geq 0 \end{cases}$$

At $x = 0$

$$f'(x) = \begin{cases} -\sin x - 1, & x < 0 \\ -\sin x + 1, & x > 0 \end{cases}$$

$$\text{LHD} = -1, \text{ RHD} = 1$$

$\therefore$ Not differentiable

$$f(x) = \cos |x| - |x| = \begin{cases} \cos x + x & , x < 0 \\ \cos x - x & , x \geq 0 \end{cases}$$

Not differentiable at $x = 0$

$$f(x) = \sin |x| + |x| = \begin{cases} -\sin x - x & , x < 0 \\ +\sin x + x & , x \geq 0 \end{cases}$$

Not differentiable at $x = 0$

$$f(x) = \sin |x| - |x| = \begin{cases} -\sin x + x & , x < 0 \\ +\sin x - x & , x \geq 0 \end{cases}$$

At $x = 0$

$$f'(x) = \begin{cases} -\cos x + 1 & , x < 0 \\ +\cos x - 1 & , x \geq 0 \end{cases}$$

LHD = 0, RHD = 0

$\therefore f$ is differentiable at $x = 0$

23. Let $h(x) = |x|$, then $g(x) = |f(x)| = h(f(x))$
 Since, composition of two continuous functions is continuous, g is continuous if f is continuous.
 So answer is (C).

(A) is wrong answer. Let $f(x) = x \Rightarrow g(x) = |x|$
 Now, $f(x)$ is an onto function. Since co-domain of x is $\mathbb{R}$ and range of x is $\mathbb{R}$. But $g(x)$ is into function. Since range of $g(x)$ is $[0, \infty)$ but co-domain is given $\mathbb{R}$.

(B) Let $f(x) = x \Rightarrow g(x) = |x|$. Now $f(x)$ is one-one function but $g(x)$ is many-one function.
 Hence (B) is wrong.

(D) Let $f(x) = x \Rightarrow g(x) = |x|$. Now, $f(x)$ is differentiable for all $x \in \mathbb{R}$ but $g(x) = |x|$ is not differentiable at $x = 0$. Hence (D) is wrong.

24. We have, for $x \neq 0$,

$$f(x) = \frac{x}{1+x} + \frac{x}{(x+1)(2x+1)} + \frac{x}{(2x+1)(3x+1)} + \dots \infty$$

$$= \sum_{n=1}^{\infty} \frac{x}{[(n-1)x+1][nx+1]}$$

$$= \sum_{n=1}^{\infty} \left[\frac{1}{(n-1)x+1} - \frac{1}{nx+1} \right] = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{nx+1} \right] = 1$$

$$\text{for } x = 0, f(x) = 0. \text{ Thus, } f(x) = \begin{cases} 1, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\text{clearly, } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = 1 \neq f(0)$$

so, $f(x)$ is discontinuous at $x = 0$

Hence, (B) is correct answer

25. The given function is

$$f(x) = \begin{cases} \tan^{-1} x & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1) & \text{if } |x| > 1 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} \frac{1}{2}(-x - 1) & \text{if } x < -1 \\ \tan^{-1} x & \text{if } -1 \leq x \leq 1 \\ \frac{1}{2}(x - 1) & \text{if } x > 1 \end{cases}$$

Clearly L.H.L. at $(x = -1) = \lim_{h \rightarrow 0} f(-1 - h)$

R.H.L. at $(x = -1) = \lim_{h \rightarrow 0} f(-1 + h)$

$$= \lim_{h \rightarrow 0} \tan^{-1}(-1 + h) = -\pi/4$$

$\therefore$ L.H.L. $\neq$ R.H.L. at $x = -1$

$\therefore f(x)$ is discontinuous at $x = -1$

Also we can prove in the same way, that $f(x)$ is discontinuous at $x = 1$

$\therefore f(x)$ can not be found for $x = \pm 1$ or domain of $f'(x) = \mathbb{R} - \{-1, 1\}$

26. $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$
 $[\because f'(a) > 0, \text{ being strictly increasing}]$
 Using L.H. Rule

$$\lim_{x \rightarrow 0} \frac{f'(x^2) \cdot 2x - f'(x)}{f'(x)} = \lim_{x \rightarrow 0} \frac{f'(x^2) \cdot 2x}{f'(x)} - 1 = 0 - 1 = -1$$

27. Given function is $y = ||x| - 1|$ or

$$y = \begin{cases} -|x| + 1 & \text{if } |x| < 1 \\ |x| - 1 & \text{if } |x| \geq 1 \end{cases}$$

$$= \begin{cases} -|x| + 1 & \text{if } -1 < x < 1 \\ |x| - 1 & \text{if } x \leq -1 \text{ or } x \geq 1 \end{cases}$$

Here $Ly'(-1) = -1$ and $Ry'(-1) = 1$
 $Ly'(0) = 1$ and $Ry'(0) = -1$ and $Ly'(1) = -1$
 and $Ry'(1) = 1$
 $\Rightarrow y$ is not differentiable at $x = -1, 0, 1 \therefore$

$$28. \quad f(0) = f(0^+) = \lim_{n \rightarrow \infty} f\left(\frac{1}{n}\right) = 0$$

$$f'(0) = \lim_{x \rightarrow 0}$$

$$\frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{n \rightarrow \infty} \frac{f(1/n)}{1/n} = 0$$

$$29. \quad y \cos x + x \cos y = \pi$$

$$\Rightarrow \frac{dy}{dx} = \frac{x \sin y - \cos x}{\cos y - y \sin x}$$

$$\text{When } x = 0, \frac{dy}{dx} = 1$$

$$(\cos y - y \sin x) \left[x \cos y \frac{dy}{dx} + \sin y + \sin x \right] - (x \sin y - \cos x)$$

$$\frac{d^2 y}{dx^2} = \frac{[-\sin y \frac{dy}{dx} - y \cos x - \sin x \frac{dy}{dx}]}{(\cos y - y \sin x)^2}$$

$$\Rightarrow \left(\frac{d^2 y}{dx^2} \right)_{x=0} = \pi.$$

$$30. \quad f''(x) = -f(x) \Rightarrow \frac{d}{dx} f'(x) = -f(x)$$

$$g'(x) = -f(x) \text{ \& } f'(x) = g(x)$$

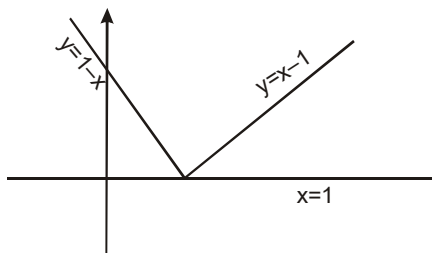
$$F(x) = \left(f\left(\frac{x}{2}\right) \right)^2 + \left(g\left(\frac{x}{2}\right) \right)^2$$

$$F'(x) = 0 \Rightarrow F(x) = C \Rightarrow F(10) = 5$$

$$0 + e^{\lim_{x \rightarrow 0} \sin x \ln\left(\frac{1}{x}\right)} = 1.$$

31. Use sandwich Theorem.

32.



According to question we get, $\lim_{x \rightarrow 1^+} g(x) = -1$

From graph : $p = -1$

$$g(x) = \frac{(x-1)^n}{\log(\cos^m(x-1))} \quad 0 < x < 2, m, n \in \mathbb{N}$$

$$g(x) = \lim_{x \rightarrow 1^+} \frac{(x-1)^n}{\log \cos^m(x-1)} = -1$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$g(1+) = \lim_{h \rightarrow 0} \frac{(h)^n}{\log \cos^m h}, \quad h > 0$$

$$= \lim_{h \rightarrow 0} \frac{h^n}{m(\ln \cosh)} = \frac{1}{m} \lim_{h \rightarrow 0} \frac{h^n}{(\ln \cosh)}$$

$$= \frac{1}{m} \lim_{h \rightarrow 0} \frac{n h^{n-1}}{-\sinh} \times \cosh = \frac{-n}{m} \lim_{h \rightarrow 0} \frac{h^{n-2}}{\left(\frac{\sinh}{h}\right)} \times (\cosh)$$

$$\Rightarrow n = 2, m = 2$$

$$33. \quad \text{Let } g(x) = -\sqrt{9-x^2} \text{ is defined for } x \in [-3, 3]$$

so, $f(x)$ is defined on $[-3, 3]$

It is clear from the graph that $f(x)$ is continuous but not differentiable at A, B and C. It is note that at point P, right hand derivative $-\infty$ and Q, left hand derivative is $+\infty$.

So, $f(x)$ is not differentiable at P and Q.

$$34. \quad \text{Let } g(x) = x^2 + (k-1)x - k$$

$$\Rightarrow g(|x|) = x^2 + (k-1)|x| - k$$

$g(x)$ must have two distinct roots with same sign

$$\Rightarrow -k > 0 \Rightarrow k < 0$$

$$35. \quad \text{For } x > 0$$

$$f(x) = \lim_{n \rightarrow \infty} \left(\frac{\frac{ax^2 + bx + c}{e^{nx}} + 1}{\frac{1}{e^{nx}} + c} \right) = \frac{0+1}{0+c} = \frac{1}{c}$$

For $x = 0$

$$f(x) = \lim_{n \rightarrow \infty} \frac{0+0+c+1}{1+c} = 1$$

and for $x < 0$

$$f(x) = \lim_{n \rightarrow \infty} \left(\frac{ax^2 + bx + c + e^{nx}}{1 + ce^{nx}} \right) = ax^2 + bx + c$$

$$\Rightarrow f(x) = \begin{cases} \frac{1}{c} & , x > 0 \\ 1 & , x = 0 \\ ax^2 + bx + c & , x < 0 \end{cases}$$

Since $f(x)$ is continuous everywhere.

$$\therefore f(0^+) = f(0) = f(0^-)$$

$$\Rightarrow \frac{1}{c} = 1 = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} (ah^2 - bh + c) = c$$

$$\Rightarrow c = 1, a, b \in R$$

36. by choosing any arcs of circle $x^2 + y^2 = 4$, we can define a discontinuous function, one of which is

$$f(x) = \sqrt{4 - x^2}, -2 \leq x < 0$$

$$= -\sqrt{4 - x^2}, 0 \leq x \leq 2$$

37. We have $Lf'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$

$$= \lim_{h \rightarrow 0} \frac{(1-h)[1-h] - 1[1]}{-h} = \lim_{h \rightarrow 0} \frac{0-1}{-h} \rightarrow \infty$$

$f'(1)$ does not exist.

Also, $Lf'(2) = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h}$

$$= \lim_{h \rightarrow 0} \frac{(2-h)[2-h] - (2-1)[2]}{-h} = \lim_{h \rightarrow 0} \frac{2-h-1}{-h} = 1$$

and $Rf'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$

$$= \lim_{h \rightarrow 0} \frac{(2+h)[2+h] - (2-1)[2]}{h} = \lim_{h \rightarrow 0} \frac{2(1+h)-2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2h}{h} \rightarrow 2$$

$f'(2)$ also does not exist

Hence, (A) is correct answer

38. $f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$

$$= \lim_{h \rightarrow 0} \frac{h^2 \left| \cos \frac{\pi}{h} \right| - 0}{h} = \lim_{h \rightarrow 0} h \cos \left(\frac{\pi}{h} \right) = 0$$

so, $f(x)$ is differentiable at $x = 0$

$$f'(2^+) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(2+h)^2 \cos \left(\frac{\pi}{2+h} \right)}{h} = \pi$$

Again, $f'(2^-) = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h}$

$$= \lim_{h \rightarrow 0} \frac{(2-h)^2 \left| \cos \left(\frac{\pi}{2-h} \right) \right|}{-h} = -\pi$$

39. At $x = 1$, L.H.L. = 1, R.H.L. = 1 $f(1) = 1$,
So $f(x)$ is continuous at $x = 1$ L.H.D. = 0,
R.H.D. = 2, so $f(x)$ is not differentiable at $x = 1$

MULTI ANSWER QUESTIONS

40. $\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0+h)$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h)}{(1-h)} \lim_{h \rightarrow 0} \frac{\cos^{-1}(1-h)}{\sqrt{2}h} = \infty$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1+h-1) \cdot \cos^{-1}(1+h-1)}{\sqrt{2}(-h+1) \cdot (1+h-1)}$$

$$= 1 \cdot \frac{\pi/2}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}}$$

41. Using expansions, given limit =

$$\lim_{x \rightarrow 0} \frac{(a-b)x + cx^2 + \left(1 - \frac{a}{3!}\right)x^3 + \frac{a}{5!}x^5 - \dots}{\frac{2}{3}x^5 - \frac{1}{2}x^6}$$

which is finite

$$\therefore a-b=0, c=0, \left(1 - \frac{a}{3!}\right) = 0$$

42. $(gof)(x) = g(f(x)) = g([x]) = 0$

Hence $(gof)(x) = 0 = \text{constant}$.

$$(fog)(x) = \begin{cases} [0], & x \in I \\ [x^2], & x \notin I \end{cases} = \begin{cases} 0, & x \in I \\ [x^2], & x \notin I \end{cases}$$

Clearly, $(fog)(x)$ is continuous at $x = 0$ and discontinuous at all other integral points.

43. (A) $\lim_{x \rightarrow 1^+} \frac{1}{\ln|x|} = \infty$ and $\lim_{x \rightarrow 1^-} \frac{1}{\ln|x|} = -\infty$

hence $f(x)$ has non removable discontinuity.

$$(B) \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1} = \frac{2}{3}$$

$f(x)$ has removable discontinuity at $x = 1$

$$(C) \lim_{x \rightarrow 1^+} \left(2^{-\frac{1}{1-x}} \right) = 1 \text{ and } \lim_{x \rightarrow 1^-} \left(2^{-\frac{1}{1-x}} \right) = 0$$

Hence, the limit does not exist.

$$(D) \lim_{x \rightarrow 1} \frac{\sqrt{x+1} - \sqrt{2x}}{x^2 - x} = \frac{-1}{2\sqrt{2}} \text{ (Rationalizing)}$$

Therefore, $f(x)$ has removable discontinuity at $x = 1$.

44. By graph of $f(x)$

$$45. \left. \frac{d}{dx} (f(x)g(x)) \right|_{x=a} = f'(a)g(a) + \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} \cdot f(a)$$

If $f(a) \neq 0 \Rightarrow g'(a)$ must exist.

Also, if $g(a)$ is discontinuous, $f(a)$ must be 0 for $f(x) \cdot g(x)$ to be differentiable.

46. $f(x)$ is continuous for all x if it is continuous at $x = 1$ for which $|1| - 3 = |1 - 2| + a \Rightarrow a = -3$

and $g(x)$ is continuous for all x if it is continuous at $x = 2$ for which

$$2 - |2| = \text{sgn}(2) - b \Rightarrow 0 = 1 - b \Rightarrow b = 1$$

Thus, $h(x) = f(x) + g(x)$ is continuous for all x if $a = -3, b = 1$

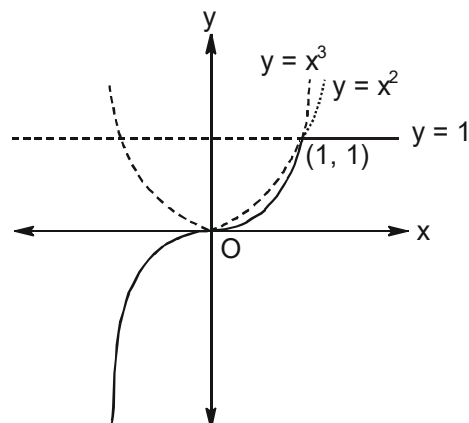
Hence, $h(x) = f(x) + g(x)$ is discontinuous at exactly one point for options (a), (b) and (c).

47. We have, for $-1 \leq x \leq 1 \Rightarrow 0 \leq x \sin \pi x \leq 1/2$
 $\therefore f(x) = [x \sin \pi x] = 0$

Also $x \sin \pi x$ becomes negative and numerically less than 1 when x is slightly greater than 1 and so by definition of $[x]$.
 $f(x) = [x \sin \pi x] = -1$ when $1 < x < 1 + h$
 thus $f(x)$ is constant and equal to 0 in the closed interval $[-1, 1]$ and so $f(x)$ is continuous and differentiable in the open interval $(-1, 1)$.
 At $x = 1$, $f(x)$ is clearly discontinuous, since $f(1 - 0) = 0$ and $f(1 + 0) = -1$ and $f(x)$ is non-differentiable at $x = 1$.

48. from graph $f(x)$ is continuous every where but not differentiate at $x = 1$.

LIMITS, CONTINUITY & DIFFERENTIABILITY



$$50. f(x) = \begin{cases} \frac{x \log \cos x}{\log(1+x^2)}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$\text{L.H.L} = \lim_{x \rightarrow 0} f(x) = \lim_{h \rightarrow 0} \frac{-h \log \cosh}{\log(1+h^2)}$$

$$= -\lim_{h \rightarrow 0} \frac{\log \cosh - \frac{h}{\cosh} \sinh}{\frac{1}{1+h^2} \times 2h}$$

$$= -\lim_{h \rightarrow 0} \frac{(\log \cosh - h \tanh)(1+h^2)}{2h}$$

$$= -\lim_{h \rightarrow 0} \frac{(-\tanh - \tanh - h \sec^2 h)(1+h^2) + 2h(\log \cosh - h \tanh)}{2}$$

$$= -\frac{1}{2}(0 - 0 - 0)(1 + 0) + 0 = 0$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} \frac{\log \cosh}{\frac{\log(1+h^2)}{h^2} \times h} = \lim_{h \rightarrow 0} \frac{\log \cosh}{h} = \lim_{h \rightarrow 0} \frac{\tanh}{1} = 0$$

$$\lim_{x \rightarrow 0} f(x) = 0, f \text{ is continuous at } x = 0$$

$$\text{L.H.D} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{-h \log \cosh}{\log(1+h^2)} - 0}{-h} = \lim_{h \rightarrow 0} \frac{\log \cosh}{\log(1+h^2)}$$

$$= \lim_{h \rightarrow 0} \frac{-\tanh(1+h^2)}{2h} = -\frac{1}{2}$$

R.H.D. $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h \log \cosh}{\log(1+h^2)}$

$$= \lim_{h \rightarrow 0} \frac{-\tanh(1+h^2)}{2h} = -\frac{1}{2}$$

$f(x)$ is differentiable at $x = 0$

51. $\lim_{x \rightarrow a} [f(x)]$ can exist only when $f(x)$ either increases or decreases at both sides of the point $x = a$. Since $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$. So this can occur only when $\lim_{x \rightarrow a} f(x)$ is an integer.

52. We have $f(x) = \frac{1}{[\sin x]}$

$f(x)$ is defined, when $-1 \leq \sin x < 0$ and $\sin x = 1$

$$\therefore x \in ((2n+1)\pi, (2n+2)\pi) \cup \left\{2n\pi + \frac{\pi}{2}\right\},$$

where $n \in I$

$\therefore f(x)$ is continuous function.

Hence, $f(x)$ is continuous in

$((2n+1)\pi, (2n+2)\pi)$, where $n \in I$.

and $Lf'(\pi/2) = \lim_{h \rightarrow 0} \frac{f(\pi/2-h) - f(\pi/2)}{-h}$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{[\sin(\pi/2-h)]} - \frac{1}{\sin \pi/2}}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{[\cosh]} - 1}{-h} = -\infty$$

Hence, $f(x)$ is differentiable at $x = \pi/2$.

53. Use $f'(x) = \lim_{x \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

54. For continuity at $x = 1$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x^2 \operatorname{sgn}[x] + \{x\}) = 1 + 0 = 1$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 \operatorname{sgn}[x] + \{x\}) = 1 \operatorname{sgn}(0) = 1$$

Also, $f(1) = 1$

Hence $f(x)$ is continuous at $x = 1$

Now for differentiability

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{(1+h)^2 \operatorname{sgn}[1+h] + \{1+h\} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 + 3h}{h} = 3$$

$$\text{and } f'(-1) = \lim_{h \rightarrow 0} \frac{(1-h)^2 \operatorname{sgn}[1-h] + \{1-h\} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 - 3h}{-h} = 3$$

$$f'(1^+) = f'(1^-)$$

Hence, $f(x)$ is differentiable at $x = 1$

Now at $x = 2$,

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x^2 \operatorname{sgn}[x] + \{x\}) = 4 \times 0 + 1$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{h \rightarrow 2^+} (\sin x + |x-3|) = 1 + \sin 2$$

Hence $LHL \neq RHL$

Hence, $f(x)$ is discontinuous at $x = 2$ and

then $f(x)$ is also non differentiable at $x = 2$

56. $f(x+y) = f(x) + f(y) + xy(x+y)$

$$f(0) = 0$$

$$\lim_{h \rightarrow 0} \frac{f(h)}{h} = -1$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) + f(h) + xh(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h)}{h} + \lim_{h \rightarrow 0} x(x+h) = -1 + x^2$$

$$f'(x) = -1 + x^2$$

$$f(x) = \frac{x^3}{3} - x + c$$

$f(x)$ is a polynomial function

$f(x)$ is twice differentiable for all $x \in \mathbb{R}$ and

$$f'(3) = 3^2 - 1 = 8.$$

57. $f'(x) = 6x + 12$

for $f(x)$ is increasing $f'(x) \geq 0 \Rightarrow x \geq -2$

Hence, $f(x)$ is increasing in $[-1, 2]$

$$\lim_{x \rightarrow 2^-} f(x) = 35, \lim_{x \rightarrow 2^+} f(x) = 35$$

$$\text{and } \lim_{x \rightarrow 2} f(x) = 35$$

Hence, f is continuous on $\{-1, 3\}$

$f'(-2) = 24$ and $f'(2) = -1$ hence $f'(2)$ does not exist

for maximum, $f(2) = 35$, $f(-1) = -10$

$$f(3) = 34$$

Hence, $f(x)$ has maximum value at $x = 2$.

58. Put $x = y = 0 \Rightarrow f(0) = 1$, Now $f'(x) =$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) + f(h) + 2xh - 1 - f(x)}{h}$$

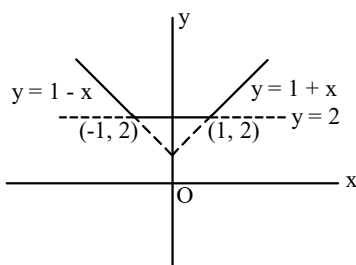
$$= 2x + \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = 2x + f'(0)$$

(since $f(0) = 1$) or $f'(x) = 2x + \sin \phi$

$$\therefore f(x) = x^2 + x \sin \phi + c \text{ but } f(0) = 1 \Rightarrow c = 1$$

$$\therefore f(x) = x^2 + x \sin \phi + 1$$

$$= \left(x + \frac{\sin \phi}{2}\right)^2 + \left(1 - \frac{\sin^2 \phi}{4}\right) \geq 1 - \frac{\sin^2 \phi}{4} \geq \frac{3}{4} > 0.$$



59.

From graph it is clear that $f(x)$ is continuous everywhere and also differentiable everywhere except at $x = 1$ and -1 .

LIMITS, CONTINUITY & DIFFERENTIABILITY

61. $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4} \quad a > 0$

$$\lim_{x \rightarrow 0} \frac{a - (a^2 - x^2)^{1/2} - \frac{x^2}{4}}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{a - a \left(1 - \left(\frac{x}{a}\right)^2\right)^{1/2} - \frac{x^2}{4}}{x^4}$$

$$\lim_{x \rightarrow 0} \frac{a - a \left(1 - \frac{1}{2} \cdot \frac{x^2}{a^2} - \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{x^4}{2a^4}\right) - \frac{x^2}{4}}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} \frac{x^2}{a} + \frac{1}{8} \frac{x^4}{a^4} - \frac{x^2}{4}}{x^4}$$

$$\frac{x^2 \left(\frac{1}{2a} - \frac{1}{4} \right) + \frac{1}{8} \frac{x^4}{a^4}}{x^4}$$

$$\text{If } \frac{1}{2a} - \frac{1}{4} = 0 \Rightarrow a = 2,$$

$$\text{if } a = 2, L = \frac{1}{8} \cdot \frac{1}{8} = \frac{1}{64}.$$

62. $\therefore f(0) = 0$ and $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{f(h)}{h} = f'(0) = k \text{ (say)}$$

$$\Rightarrow f(x) = kx + c \Rightarrow f(x) = kx \quad (\because f(0) = 0)$$

63. $\lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = 0 = f\left(-\frac{\pi}{2}\right)$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \cos\left(-\frac{\pi}{2}\right) = 0$$

$$f'(x) = \begin{cases} -1, & x \leq -\frac{\pi}{2} \\ \sin x, & -\frac{\pi}{2} < x \leq 0 \\ 1, & 0 < x \leq 1 \\ \frac{1}{x}, & x > 1 \end{cases}$$

Clearly, $f(x)$ is not differentiable at $x=0$ as

$$f'(0^-) = 0 \text{ and } f'(0^+) = 1.$$

$f(x)$ is differentiable at $x=1$ as $f'(1^-) = f'(1^+) = 1$.

64. At $x=2n$

$$\text{LHL} = \lim_{h \rightarrow 0} (b_n + \cos \pi(2n-h)) = b_n + 1$$

$$\text{RHL} = \lim_{h \rightarrow 0} (a_n + \sin \pi(2n-h)) = a_n$$

$$f(2n) = a_n$$

For continuity $b_{n+1} = a_n$

At $x=2n+1$

$$\text{LHL} = \lim_{h \rightarrow 0} (a_n + \sin \pi(2n+1-h)) = a_n + 1$$

$$\text{RHL} = \lim_{h \rightarrow 0} (b_{n+1} + \cos \pi(2n+1-h)) = b_{n+1} - 1$$

$$\lim_{h \rightarrow 0} (a_n + \sin f(2n+1)) = a_n$$

for continuity

$$a_n = b_{n+1} - 1, \quad a_{n-1} - b_n = -1.$$

COMPREHENSIONS QUESTIONS

Passage - 1: 65 TO 67

$$\text{Let } f(x) = \lim_{n \rightarrow \infty} \left(\cos \sqrt{\frac{x}{n}} \right)^n,$$

$$g(x) = \lim_{n \rightarrow \infty} (1 + x + x^n \sqrt[n]{e})^n. \text{ Now, consider}$$

the function $y = h(x)$, where

$$h(x) = \tan^{-1}(g^{-1}f^{-1}(x)). \text{ On solving we get}$$

$$f(x) = e^{\frac{x}{2}}, \quad g(x) = e^x, \quad f^{-1}(x) = \ln \frac{1}{x^2} \text{ \&}$$

$$g^{-1}(x) = \ln x$$

$$65. \quad \lim_{x \rightarrow 0^+} \frac{\ln f(x)}{\ln g(x)} = \lim_{x \rightarrow 0} \frac{-x/2}{x} = -\frac{1}{2}$$

66. Domain of $h(x)$ is $(0, 1)$

$$67. \quad h(x) = \tan^{-1} \left(\ln \left(\ln \frac{1}{x^2} \right) \right) \quad 0 < x < 1$$

$$1 < \frac{1}{x^2} < \infty \Rightarrow 0 < x < 1$$

$$-\infty < \ln \left(\ln \frac{1}{x^2} \right) < \infty$$

$$\text{Range of } h(x) \text{ is } \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

Passage - 2: 68 TO 70

$$\text{For } 0 \leq x \leq \frac{\pi}{4}, \quad g(x) = 1 + \tan x$$

$$x \in \left[0, \frac{\pi}{4} \right) \Rightarrow 1 + \tan x \in [1, 2)$$

$$\text{so } f(g(x)) = f(1 + \tan x) = 1 + \tan x + 2$$

$$\text{and for } x \in \left[\frac{\pi}{4}, \pi \right), \quad g(x) = 3 - \cot x$$

$$x \in \left[\frac{\pi}{4}, \pi \right) \Rightarrow 3 - \cot x \in [2, \infty)$$

$$\text{so } f(g(x)) = f(3 - \cot x) = 6 - (3 - \cot x)$$

Let

$$h(x) = f(g(x)) = \begin{cases} 3 + \tan x, & 0 \leq x < \frac{\pi}{4} \\ 3 + \cot x, & \frac{\pi}{4} \leq x < \pi \end{cases}$$

clearly, $f(g(x))$ is continuous in $[0, \pi)$

$$\text{Now } h' \left(\frac{\pi^+}{4} \right) = \lim_{x \rightarrow \frac{\pi^+}{4}} (-\cos \sec^2 x) = -2$$

$$h' \left(\frac{\pi^-}{4} \right) = \lim_{x \rightarrow \frac{\pi^-}{4}} (\sec^2 x) = 2$$

So $f(g(x))$ is differentiable every where in

$(0, \pi]$ other than at $x = \frac{\pi}{4}$

$$|f(g(x))| = \begin{cases} |3 + \tan x|, & 0 \leq x < \frac{\pi}{4} \\ |3 + \cot x|, & \frac{\pi}{4} \leq x < \pi \end{cases}$$

which is non differentiable at $x = \frac{\pi}{4}$ and

where $3 + \cot x = 0$ or $x = \cot^{-1}(-3)$

For $x \in \left[0, \frac{\pi}{4}\right), 3 + \tan x \in [3, 4)$

For $x \in \left[\frac{\pi}{4}, \pi\right), 3 + \cot x \in (-\infty, 4]$

Hence the range is $(-\infty, 4]$

Passage - 3: 71 TO 73

We have

$$f(x) = \frac{\sin^{-1}(1 - \{x\}) \cdot \cos^{-1}(1 - \{x\})}{\sqrt{2\{x\}}(1 - \{x\})}$$

$$1. (D) R = \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1 - \{0+h\}) \cdot \cos^{-1}(1 - \{0+h\})}{\sqrt{2\{0+h\}}(1 - \{0+h\})}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h) \cdot \cos^{-1}(1-h)}{\sqrt{2h}(1-h)}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h)}{(1-h)} \lim_{h \rightarrow 0} \frac{\cos^{-1}(1-h)}{\sqrt{2h}}$$

in second limit put $1-h = \cos \theta$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h)}{(1-h)} \lim_{\theta \rightarrow 0} \frac{\cos^{-1}(\cos \theta)}{\sqrt{2(1-\cos \theta)}}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h)}{(1-h)} \lim_{\theta \rightarrow 0} \frac{\theta}{2 \sin(\theta/2)}$$

$$= \sin^{-1} 1 \cdot 1 = \pi/2 \Rightarrow 100R = 50\pi$$

$$\therefore \cos(100R) = \cos 50\pi = (-1)^{50} = 1$$

$$2. (a): L = \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0-h)$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1 - \{0-h\}) \cdot \cos^{-1}(1 - \{0-h\})}{\sqrt{2\{0-h\}}(1 - \{0-h\})}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1+h-1) \cdot \cos^{-1}(1+h-1)}{\sqrt{2(-h+1)}(1-h-1)}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1} h}{h} \cdot \lim_{h \rightarrow 0} \frac{\cos^{-1} h}{\sqrt{2(1-h)}}$$

$$= 1 \cdot \frac{\pi/2}{\sqrt{2}} = \frac{\pi}{2\sqrt{2}} \Rightarrow 99\sqrt{2}L = \frac{99\pi}{2}$$

$$\therefore \sin(99\sqrt{2}L) = \sin\left(\frac{99\pi}{2}\right) = (-1)^{\frac{99-1}{2}} = (-1)^{49} = -1$$

$$21 (C) \therefore R = \frac{\pi}{2} \text{ and } L = \frac{\pi}{2\sqrt{2}}$$

$$\therefore 2R^2 + 4L^2 = \pi^2$$

$$\Rightarrow [2R^2 + 4L^2] = [\pi^2] = [9.87] = 9.$$

Passage - 4: 74 TO 76

$$f(x) = \begin{cases} [x], & -2 \leq x \leq -\frac{1}{2} \\ 2x^2 - 1, & -\frac{1}{2} < x \leq 2 \end{cases}$$

$$= \begin{cases} -2, & -2 \leq x \leq -1 \\ -1, & -1 \leq x \leq -\frac{1}{2} \\ 2x^2 - 1, & -\frac{1}{2} < x \leq 2 \end{cases}$$

$$|f(x)| = \begin{cases} 2, & -2 \leq x \leq -1 \\ 1, & -1 \leq x \leq -\frac{1}{2} \\ |2x^2 - 1|, & -\frac{1}{2} < x \leq 2 \end{cases}$$

$$= \begin{cases} 2, & -2 \leq x \leq -1 \\ 1, & -1 \leq x \leq -\frac{1}{2} \\ 1 - 2x^2, & -\frac{1}{2} < x \leq \frac{1}{\sqrt{2}} \\ 2x^2 - 1, & \frac{1}{\sqrt{2}} < x \leq 2 \end{cases}$$

MATRIX-MATCH QUESTIONS

 77. (A): Here $a > 0$, if $a \leq 0$, then limit = ∞

$$\therefore \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 - x + 1 - ax - b})(\sqrt{x^2 - x + 1 + ax + b})}{(\sqrt{x^2 - x + 1 + ax + b})}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{(x^2 - x + 1) - (ax + b)^2}{(\sqrt{x^2 - x + 1 + ax + b})}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{(1 - a^2)x^2 - (1 + 2ab)x + (1 - b^2)}{(\sqrt{x^2 - x + 1 + ax + b})}$$

 This is possible only when $1 - a^2 = 0$ and

$$1 + 2ab = 0$$

$$\therefore a = \pm 1$$

$$\Rightarrow a = 1 \quad (\because a > 0) \dots (1)$$

$$\Rightarrow b = -1/2$$

$$\Rightarrow (a, 2b) = (1, -1)$$

 (B): Divide numerator and denominator by $e^{1/x}$, then

$$\lim_{x \rightarrow \infty} \frac{(1 + a^3)e^{1/x} + 8}{e^{1/x} + (1 - b^3)} = 2$$

$$\Rightarrow \frac{0 + 8}{0 + 1 - b^3} = 2 \Rightarrow 1 - b^3 = 4$$

$$\therefore b^3 = -3 \rightarrow b = -3^{1/3}$$

$$\text{Then } a \in \mathbb{R} \Rightarrow (a, b^3) = (a, -3)$$

$$(C): \lim_{x \rightarrow \infty} (\sqrt{(x^4 - x^2 + 1)} - ax^3 - b) = 0$$

$$\text{Put } x = \frac{1}{t},$$

$$\therefore \lim_{x \rightarrow 0} \left(\sqrt{\left(\frac{1}{t^4} - \frac{1}{t^2} + 1\right)} - \frac{a}{t^2} - b \right) = 0$$

$$\Rightarrow \lim_{t \rightarrow 0} \frac{\sqrt{(1 - t^2 + t^4)} - a - bt^2}{t^2} = 0 \dots (1)$$

 Since RHS is finite, numerator must be equal to 0 at $t \rightarrow 0$.

$$\therefore 1 - a = 0 \therefore a = 1$$

$$f(|x|) = \begin{cases} -2, & -2 < |x| < -1 \\ -1, & -1 \leq |x| < -\frac{1}{2} \\ 2|x|^2 - 1, & -\frac{1}{2} < |x| \leq 2 \end{cases}$$

$$= 2x^2 - 1, -2 \leq x \leq 2$$

$$\Rightarrow g(x) = f(|x|) + |f(x)| = \begin{cases} 2x^2 + 1, & -2 \leq x < -1 \\ 2x^2, & -1 \leq x \leq -\frac{1}{2} \\ 0, & -\frac{1}{2} < x < \frac{1}{\sqrt{2}} \\ 4x^2 - 2, & \frac{1}{\sqrt{2}} \leq x \leq 2 \end{cases}$$

$$g(-1^-) = \lim_{x \rightarrow -1} (2x^2 + 1) = 3,$$

$$g(-1^+) = \lim_{x \rightarrow -1} 2x^2 = 2$$

$$g\left(-\frac{1}{2}\right) = \lim_{x \rightarrow -\frac{1}{2}} 2x^2 = \frac{1}{2},$$

$$g\left(-\frac{1}{2}^+\right) = \lim_{x \rightarrow -\frac{1}{2}} 0 = 0$$

$$g\left(\frac{1}{\sqrt{2}}\right) = \lim_{x \rightarrow \frac{1}{\sqrt{2}}} 0 = 0,$$

$$g\left(\frac{1}{\sqrt{2}}^+\right) = \lim_{x \rightarrow \frac{1}{\sqrt{2}}} (4x^2 - 2) = 0$$

 Hence, $g(x)$ is discontinuous at $x = -1, -\frac{1}{2}$.

$$g(x) \text{ is continuous at } x = \frac{1}{\sqrt{2}}$$

$$\text{Now, } g'\left(\frac{1}{\sqrt{2}}\right) = 0,$$

$$g'\left(\frac{1}{\sqrt{2}}^+\right) = 8\left(\frac{1}{\sqrt{2}}\right) = \frac{8}{\sqrt{2}}$$

 Hence, $g(x)$ is non-differentiable at $x = \frac{1}{\sqrt{2}}$.

from equation (1),

$$\lim_{t \rightarrow 0} \frac{\sqrt{(1-t^2+t^4)} - 1 - bt^2}{t^2} = 0$$

$$\lim_{t \rightarrow 0} (-1+t^2) \left(\frac{(1-t^2+t^4)^{1/2} - (1)^{1/2}}{(1-t^2+t^4) - 1} \right) = b$$

$$\Rightarrow (-1) \left(\frac{1}{2} \right) = b \Rightarrow a = 1, b = -\frac{1}{2}$$

$$\Rightarrow (a, -2b) = (1, 2)$$

$$(D): \lim_{x \rightarrow -a} \frac{a^7 - (-x)^7}{a - (-x)} = 7 \Rightarrow 7a^6 = 7$$

$$\Rightarrow a^6 = 1 \Rightarrow a = -1$$

78. Conceptual

79. Conceptual

$$80. (A) f(x) = \lim_{n \rightarrow \infty} [\cos^2(2\pi x)]^n + \left\{ x + \frac{1}{2} \right\}$$

$$\text{obviously, } \lim_{x \rightarrow \frac{1}{2}^+} f(x) = 0 + 0 = 0$$

$$\text{and } \lim_{x \rightarrow \frac{1}{2}^-} f(x) = 0 + 1$$

Hence, $f(x)$ is discontinuous at $x = \frac{1}{2}$

$$(B) f(x) = (\log x)(x-1)^{1/5}$$

obviously, $f(x)$ is continuous at $x = 1$

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{\log(1+h)h^{1/5}}{h} = 0$$

$$f'(1^-) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\log(1-h)(-h)^{1/5}}{-h} = 0$$

Hence, $f(x)$ is differentiable at $x = 1$

(C)

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} [\cos 2\pi x] + \lim_{x \rightarrow 1^-} \sqrt{\sin\left(\frac{\pi x}{2}\right)}$$

$$= 0 + 0 = 1$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} [\cos 2\pi x] + \lim_{x \rightarrow 1^+} \sqrt{\sin\left(\frac{\pi x}{2}\right)}$$

$$= 0 + 1 = 1, \text{ Also } f(1) = 1 + 0 = 1.$$

$f(x)$ is continuous at $x = 1$

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{[\cos 2\pi(1+h)] + \sqrt{\sin\left(\frac{\pi(1+h)}{2}\right)} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{\cos\left(\frac{\pi h}{2}\right)} - 1}{h} = 0, \text{ similarly, } f'(1^-) = 0$$

$f(x)$ is continuous when $\cos 2x = \sin x$

which has $x = \frac{\pi}{6}$ as one of the solution.

Hence it is continuous.

Also in the neighbourhood of $x = \frac{\pi}{6}$,

$$f'(x) = \begin{cases} -2 \sin 2x, & \frac{\pi}{6} - \delta < x < \frac{\pi}{6} \\ \cos x, & \frac{\pi}{6} < x < \frac{\pi}{6} + \delta \end{cases}$$

$$\text{Here, } f'(x) = \begin{cases} -2 \sin 2x, & \frac{\pi}{6} - \delta < x < \frac{\pi}{6} \\ \cos x, & \frac{\pi}{6} < x < \frac{\pi}{6} + \delta \end{cases}$$

$\Rightarrow f(x)$ is not differentiable at $x = \frac{\pi}{6}$.

81. Conceptual

82. Conceptual

83. (A) $f(x) = x|x| = \begin{cases} -x^2, & x < 0 \\ x^2, & x > 0 \end{cases}$ is continuous and differentiable everywhere. also increasing.

$$(B) f(x) = \sqrt{|x|} = \begin{cases} \sqrt{-x}, & x < 0 \\ \sqrt{x}, & x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} -\frac{1}{2\sqrt{-x}}, & x < 0 \\ \frac{1}{2\sqrt{x}}, & x > 0 \end{cases}$$

Continuous everywhere differentiable everywhere except at $x = 0$. Not increasing.

(C) $f(x) = x + [x]$ At integral point $x = I$,
LHL = $I + (I - 1) = 2I - 1$, RHL = $I + I = 2I$
= $f(I)$, So not continuous hence not differentiable at integral points but increasing.

$$(D) f(x) = |x - 1| + |x + 1| = \begin{cases} -2x, & x < -1 \\ 2, & -1 \leq x < 1 \\ 2x, & 1 \leq x \end{cases}$$

Continuous everywhere, differentiable everywhere but not increasing in $(1, 1)$.

FILL IN BLANKS QUESTIONS

84. Conceptual

85. $f(x) = \ln(x-1) + \ln(x-2)$

$$+ \ln(x-3) + \dots + \ln(x-n)$$

$$\Rightarrow \frac{f^1(x)}{f(x)} = \frac{1}{x-1} = \frac{1}{x-2} + \frac{1}{x-3} + \dots + \frac{1}{x-n}$$

$$\Rightarrow f^1(x) = (x-2)(x-3)\dots(x-n)$$

$$+ (x-1)(x-3)\dots(x-n)$$

$$= (x-1)(x-2)\dots(x-n+1)$$

$$f^1(n) = (n-1)! = 5040 \Rightarrow n = 8$$

86. Given $f(xy) = xf(y) + yf(x)$

Differentiating both sides w.r.t. x treating y as constant, $f'(xy) \cdot y = f(y) + yf'(x)$

Putting $y = x$ and $x = 1$, then

$$f'(x) \cdot x = f(x) + xf'(1)$$

$$\Rightarrow \frac{xf'(x) - f(x)}{x^2} = \frac{f'(1)}{x}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{f(x)}{x} \right) = \frac{f'(1)}{x}$$

Integrating both sides w.r.t. x taking limits 1 to x ,

$$\frac{f(x)}{x} - \frac{f(1)}{1} = f'(1) \{ \ln x - \ln 1 \}$$

$$\Rightarrow \frac{f(x)}{x} - 0 = f'(1) \ln x \quad (\because f(1) = 0)$$

$$\text{Hence, } f(x) = f'(1)(x \ln x)$$

ASSERTION-REASONING QUESTIONS

87. $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

$$\therefore Rf'(0) = f'(0^+) = \lim_{h \rightarrow 0} \frac{\frac{h}{1+e^{1/h}} - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{1+e^{1/h}} = 0$$

$$Lf'(0) = f'(0^-) = \lim_{h \rightarrow 0} \frac{\frac{-h}{1+e^{-1/h}} - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{1+\frac{1}{e^{1/h}}} = \frac{1}{1+0} = 1 \therefore f'(0^+) = 0 \text{ and}$$

$$f'(0^-) = 1$$

88. $f(x) = \begin{cases} 3-x^2, & x > 1 \\ x^3+1, & x \leq 1 \end{cases}$

$$\therefore f'(x) = 3x^2 = 3 \text{ at } x = 1^-$$

$$f'(x) = -2x = -2 \text{ at } x = 1^+$$

89. Let $f(x) = \frac{e^{1/x} - 1}{e^{1/x} + 1}$

$$\therefore \text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0-h)$$

$$= \lim_{h \rightarrow 0} \frac{e^{-1/h} - 1}{e^{-1/h} + 1} = \frac{e^{-\infty} - 1}{e^{-1} + 1} = \frac{0 - 1}{0 + 1} = -1$$

$$\text{and R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} \frac{e^{1/h} - 1}{e^{1/h} + 1} \quad \left(\text{form } \frac{\infty}{\infty}, e^{\infty} = \infty \right)$$

(Divide above and below by $e^{1/h}$)

$$\therefore \text{R.H.L.} = \lim_{h \rightarrow 0} \frac{1 - e^{-1/h}}{1 + e^{-1/h}} = \frac{1 - e^{-\infty}}{e^{-\infty}} = \frac{1 - 0}{1 + 0} = 1$$

$\therefore \text{L.H.L.} \neq \text{R.H.L.}$

Hence $\lim_{x \rightarrow 0} \left(\frac{e^{1/x} - 1}{e^{1/x} + 1} \right)$ does not exist.

90. $f(x) = (|x| - |x - 1|)^2$

$$\therefore f(x) = \begin{cases} (-x + x - 1)^2 = 1 & , \quad x < 0 \\ (x + x - 1)^2 = (2x - 1)^2 & , \quad 0 \leq x \leq 1 \\ (x - x + 1)^2 = 1 & , \quad x > 1 \end{cases}$$

$$\therefore f'(x) = 0 \Rightarrow f'(0^-) = -4$$

$$f'(x) = 2(2x - 1) \cdot 2 = -4 \Rightarrow f'(0^+) = 0$$

$$f'(x) = 4(2x - 1) = 4 \Rightarrow f'(1^-) = 4$$

$$f'(x) \Rightarrow f'(1^+) = 0$$

91. $f(x) = g(x) \sin x$

$$f''(x) = g''(x) \sin x + 2g'(x) \cos x + g(x) \sin x$$

$$f''(0) = 2g'(0) = 0$$

$$f'(x) = g'(x) \sin x + g(x) \cos x$$

$$f'(0) = g(0)$$

For statement 1:

$$\lim_{x \rightarrow 0} \frac{g(x) \cos x - g(x)}{\sin x} = \lim_{x \rightarrow 0} \frac{g'(x) \cos x - g(x) \sin x}{\cos x}$$

$$= g'(0) = 0 = f^2(0)$$

For statement 2, $f(0) = g(0)$.

92. Let $h(x) = (f(a) - f(x))(g(b) - g(x)) \cdot e^x$

$$h(a) = h(b) = 0$$

$$\exists c \in (a, b) \text{ in such a way that } h'(c) = 0$$

$$h'(x) = -f'(x)[g(b) - g(x)e^x] - g'(x)f(x) - f(x)e^x + (f(a) - f(x))(g(b) - g(x))e^x = 0$$

$$\text{or } \frac{f'(x)}{f(a) - f(x)} + \frac{g'(x)}{g(b) - g(x)} = 1$$

93. Statement 1 is correct as though

$$|4x^2 - 12x + 5| \text{ is non differentiable at}$$

$$x = \frac{1}{2}, \frac{5}{2} \text{ but } \cos \pi x = 0 \text{ those points.}$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

so, $f'\left(\frac{1}{2}\right)$ and $f'\left(\frac{5}{2}\right)$ exists.

94. Because maxima and minima is also dependent of $f(a)$

95. Let $h(x) = f(x) - \sin x$ for $x \in \left[0, \frac{\pi}{2}\right]$

$$\Rightarrow h(0) = h\left(\frac{\pi}{2}\right) = 0$$

According to Rolles theorem for at least one

$$c \in \left(0, \frac{\pi}{2}\right), h'(c) = f'(c) - \sin c = 0$$

$$\text{for } x \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right], h\left(\frac{\pi}{2}\right) = h\left(\frac{5\pi}{2}\right) = 0$$

According to rolles theorem for at least one

$$d \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right), h'(d) = f'(d) - \cos d = 0$$

$$\text{for } x \in [c, d], h'(c) = h'(d) = 0$$

According to rolles theorem for at least one

$$x \in (c, d), h'(x) = f''(x) + \sin x = 0$$

$$\Rightarrow |f''(x)| \leq 1 \text{ for at least one } x \in \left(0, \frac{3\pi}{2}\right).$$

INTEGER QUESTIONS

96. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{\{f(x) + f(h) - 2xh - 1\} - f(x)}{h}$$

$$(\text{Using the given relation}) = \lim_{h \rightarrow 0} -2x + \lim_{h \rightarrow 0} \frac{f(h) - 1}{h}$$

$$= \lim_{h \rightarrow 0} -2x + \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

[Putting $x = 0 = y$ in the given relation we find $f(0) = f(0) + f(0) + 0 - 1 \Rightarrow f(0) = 1$]

$$\therefore f'(x) = -2x + f'(0)$$

$$\Rightarrow f(x) = -x^2 - \sin \alpha \cdot x + C$$

$$f(0) = -0 - 0 + C \Rightarrow C = 1$$

$$\therefore f(x) = -x^2 - \sin \alpha \cdot x + 1$$

$$\text{So, } f\{f'(0)\} = f(-\sin \alpha) = -\sin^2 \alpha + \sin^2 \alpha + 1$$

$$\therefore f\{f'(0)\} = 1$$

97. Given that $f(x)$ is a function satisfying

$$f(-x) = f(x), \forall x \in \mathbb{R} \quad \dots (1)$$

$$\text{Also } f'(0) \text{ exists} \Rightarrow f'(0) = Rf'(0) = Lf'(0)$$

$$\text{Now, } Rf'(0) = f'(0) \Rightarrow$$

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = f'(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = f'(0) \quad \dots (2)$$

$$\text{Again } Lf'(0) = f'(0) \Rightarrow$$

$$\lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = f'(0) \Rightarrow$$

$$\lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h} = f'(0) \Rightarrow \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = -f'(0) \dots (3)$$

[Using eq. (1)]

from equations (2) and (3) we get, $\Rightarrow f'(0) = 0$

$$98. f(1/4^n) = (\sin e^n) e^{-n^2} + \frac{n^2}{n^2 + 1}$$

$$\lim_{n \rightarrow \infty} f\left(\frac{1}{4^n}\right) = \lim_{n \rightarrow \infty} \left(\frac{\sin e^n}{e^{n^2}} + \frac{n^2}{1+n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{\sin e^n}{e^{n^2}} + \frac{1}{1 + \frac{1}{n^2}} \right), \therefore f(0) = 0 + 1 = 1$$

99. $t^2 f(x) = -2t f'(x) + f''(x) = 0$ has equal roots

$$\text{Discriminant} = 4(f'(x))^2 - 4 f(x) f''(x) = 0$$

$$\frac{f''(x)}{f'(x)} = \frac{f'(x)}{f(x)}$$

$$\ln(f'(x)) = \ln f(x) - \ln c \Rightarrow f(x) = c f'(x)$$

$$f(0) = c f'(0) \Rightarrow c = \frac{1}{2}$$

$$\frac{f'(x)}{f(x)} = 2 \Rightarrow \ln f(x) = 2x + k \Rightarrow \ln f(0) = k$$

$$\Rightarrow k = 0 \Rightarrow \ln f(x) = 2x$$

$$\therefore f(x) = e^{2x}$$

$$t^2 e^{2x} - 4te^{2x} + 4e^{2x} = 0 \Rightarrow t^2 - 4t + 4 = 0$$

$$t = 2$$

$$\left(\therefore \lim_{x \rightarrow 0} \frac{f(x) - 1}{x} - t/2 \right) = \left(\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{2x} \times 2 - \frac{2}{2} \right)$$

$$= 2 - 1 = 1$$

$$100. L = \lim_{t \rightarrow 0} \frac{t + \ln(\sqrt{1+t^2} - t)}{t^3}$$

$$= \lim_{t \rightarrow 0} \frac{1 + \frac{1}{\sqrt{1+t^2} - t} \left(\frac{t}{\sqrt{1+t^2}} - 1 \right)}{3t^2}$$

$$= \lim_{t \rightarrow 0} \frac{1 - \frac{1}{\sqrt{1+t^2}}}{3t^2} = \lim_{t \rightarrow 0} \frac{\sqrt{1+t^2} - 1}{3t^2 \sqrt{1+t^2}}$$

$$= \lim_{t \rightarrow 0} \frac{(1+t^2) - 1}{3t^2} \times \frac{1}{(\sqrt{1+t^2} + 1)}$$

$$= \lim_{t \rightarrow 0} \frac{1}{3} \frac{1}{(\sqrt{1+t^2} + 1)} = \frac{1}{6}, 6L + 5 = 6.$$

$$101. f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{2f(x) + xf(h) + h\sqrt{f(x)} - 2f(x) - xf(0) - 0\sqrt{f(x)}}{h}$$

$$\text{as } f(0) = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} x \left(\frac{f(h) - f(0)}{h - 0} \right) + \sqrt{f(x)} \Rightarrow f'(x) = \sqrt{f(x)}$$

$$\Rightarrow \int \frac{f'(x)}{\sqrt{f(x)}} dx = \int dx$$

$$\Rightarrow 2\sqrt{f(x)} = x + c \Rightarrow f(x) = \frac{x^2}{4}$$

when $\alpha = 0$ area is minimum

$$\text{required minimum area} = 2 \int_0^9 2\sqrt{y} dy =$$

$$4 \left(\frac{y^{3/2}}{3/2} \right)_0^9 = 72 \text{ sq. unit.}$$

$$102. f(x) = x + \cos x + 2, f(0) = 3 \Rightarrow g(3) = 0$$

$$g(f(x)) = x \Rightarrow g'(f(x)) \cdot f'(x) = 1$$

$$\text{putting } x = 0, g'(3) \cdot f'(0) = 1,$$

$$\text{Now, } f'(x) = 1 + \sin x \Rightarrow f'(0) = 1 \Rightarrow g'(3) = 1.$$

$$103. \frac{f(x+2y)}{3} = \frac{f(x) + 2f(y)}{3}$$

$$\frac{1}{3} f'(x+2y) = \frac{f'(x)}{3} \dots (1)$$

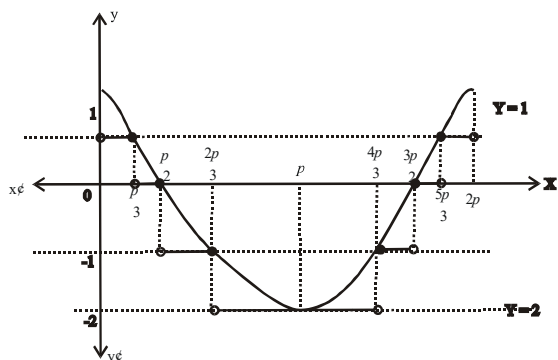
$$\frac{2}{3} f' \left(\frac{x+2y}{3} \right) = \frac{2f'(x)}{3} \dots (ii)$$

for (i & ii) $f'(x) = f'(y) \Rightarrow f'(x) = C = 1$,

$f(x) = x + d$, As $f(0) = 2$

$f(x) = x + 2$, $f(2) = 2 + 2 = 4$

104. $f(x) = [2 \cos x]$



clearly, from the graph, it can be seen that

$f(x)$ is discontinuous at $x = \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}$

105. $f(x) = \left[\frac{6x}{\pi} \right] \cos \left[\frac{3x}{\pi} \right]$

It is discontinuous when either $\frac{6x}{\pi} \in I$ or $\frac{3x}{\pi}$

$\in I$ i.e., when

$x = \pi/6$ or $\pi/3$ or $\pi/2$ or π or $2\pi/3$ or $5\pi/6$

106. $P(x) = ax^4 + bx^3 + cx^2 + dx + e$ Given

$$\lim_{x \rightarrow 0} \left(1 + \frac{P(x)}{x^2} \right) = 2$$

Limit exist only if, $d = e = 0$

$$\lim_{x \rightarrow 0} [1 + ax^2 + bx + c] = 2 \Rightarrow c + 1 = 2$$

$$\Rightarrow c = 1$$

$$P(x) = ax^4 + bx^3 + x^2$$

$$P'(x) = 4ax^3 + 3bx^2 + 2x = x(4ax^2 + 3bx + 2)$$

$$\text{Note: } 4ax^2 + 3bx + 2 \equiv \lambda(x-1)(x-2) \\ = \lambda(x^2 - 3x + 2)$$

$$\Rightarrow \lambda = 1, a = \frac{1}{4}, b = -1 \Rightarrow P(x) = \frac{1}{4}x^4 - x^3 + x^2$$

$$\therefore P(2) = \frac{1}{4}2^4 - 2^3 + 2^2 = 4 - 8 + 4 = 0$$

107. $f(x) = x^3 + e^{x/2}$, $g(x) = f^{-1}(x)$

$$f(g(x)) = x \quad f(0) = 1 \Rightarrow f^{-1}(1) = 0 \Rightarrow g(1) = 0$$

$$f'(g(x)) \cdot g'(x) = 1, \quad g'(x) = \frac{1}{f'(g(x))}$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

$$g'(1) = \frac{1}{f'(g(1))} = \frac{1}{f'(0)}$$

$$f'(x) = 3x^2 + \frac{1}{2}e^{x/2} \quad f'(0) = \frac{1}{2}. \text{ So, } g'(1) = 2.$$